Scaling Empiricism in Artificial Causal Inference

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Agenda

How to move from 'Rationalism' to 'Empiricism' in Causal Inference?

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1. Treatment Effect Exploration *(what)*

   Exploratory Causal Inference in SAEnce — Mencattini, Cadei*, Locatello*
   ICLR'26 — Oral (top ~1%)

2. Heterogeneous Treatment Effect Exploration *(how)*

   From tokens to policy: Causal and Interpretable Heterogeneous Treatment Effect Identification
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   preprint (short version at workshop AISTATS'26)

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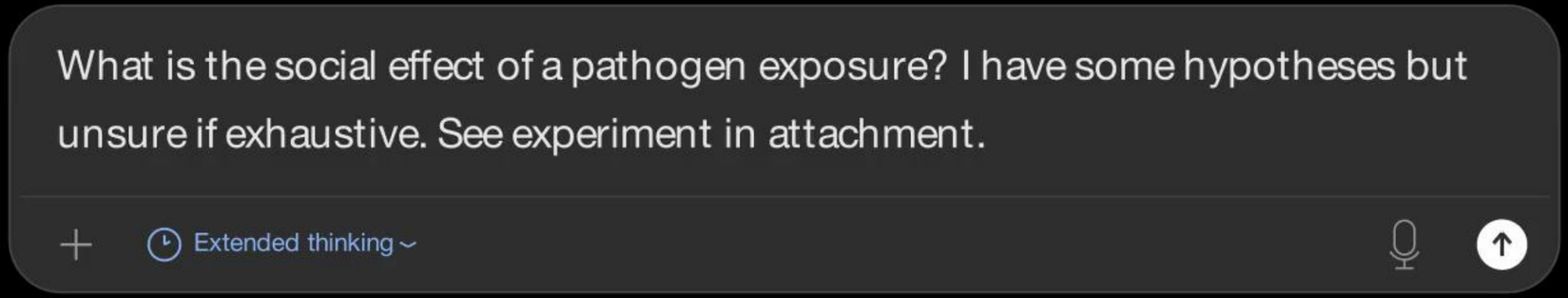
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treatment

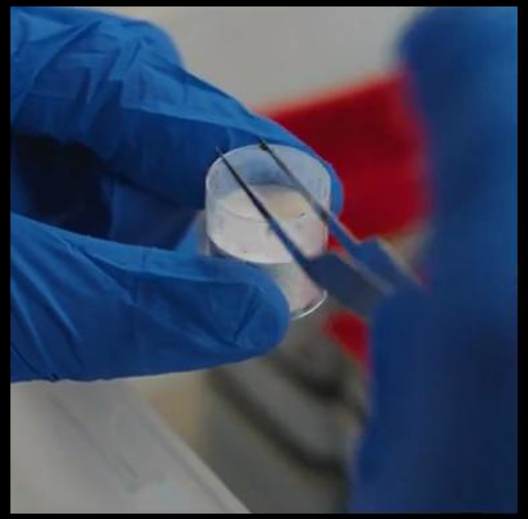


post-treatment observation

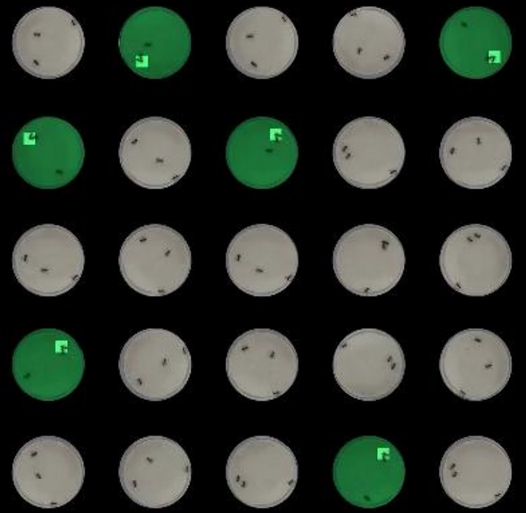
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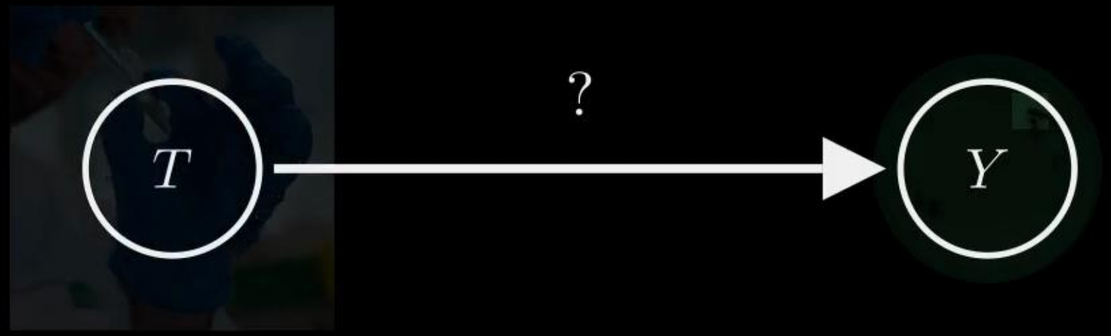


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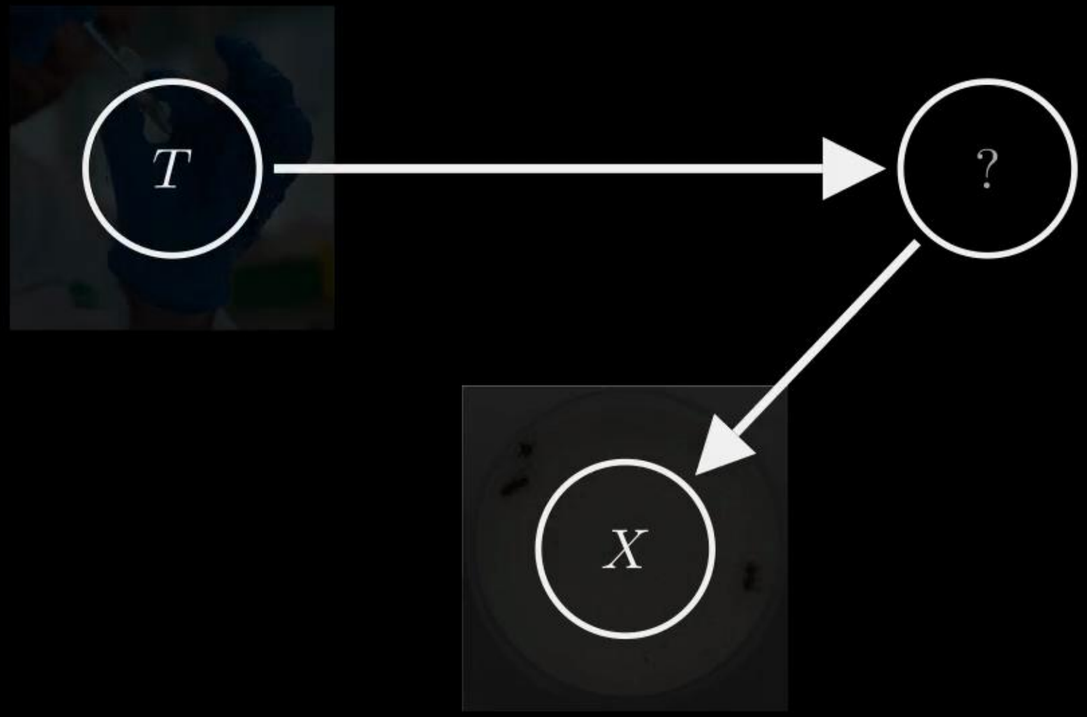


post-treatment observation

Exploratory Causal Inference



Exploratory Causal Inference



Tool: *Dictionary Learning*

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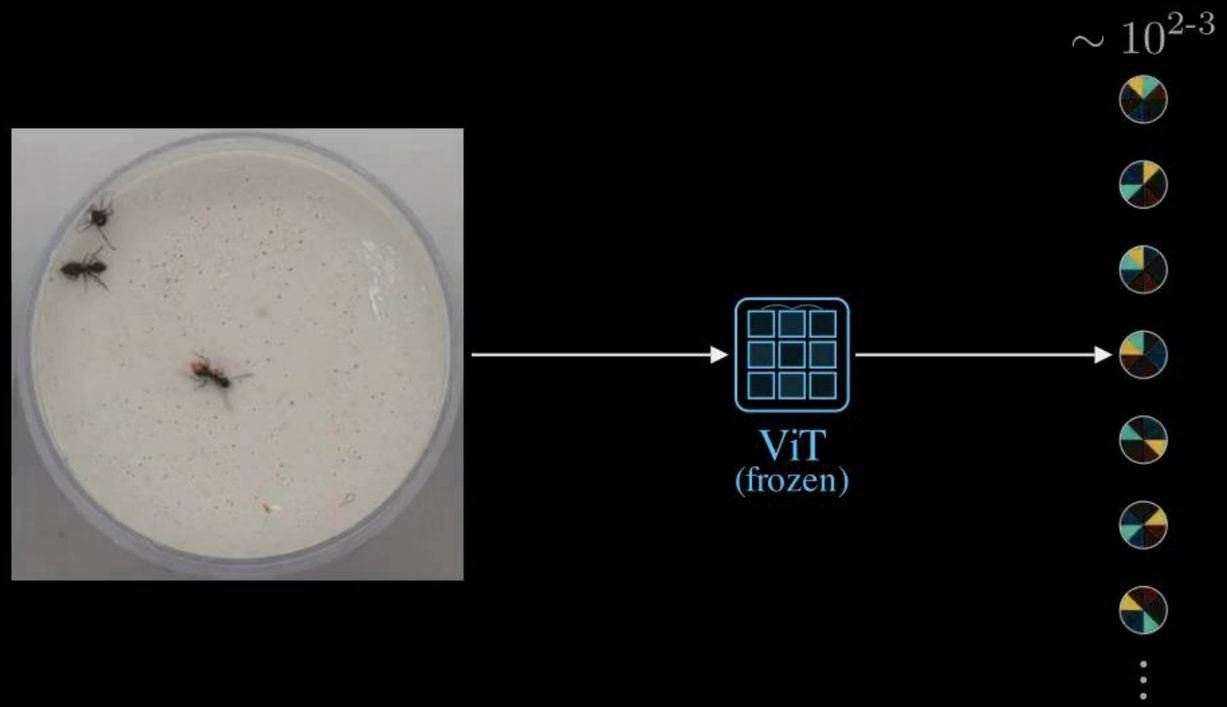
per observation:



post-treatment observation
(video)

Tool: *Dictionary Learning*

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post-treatment observation
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entangled
representation

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per observation:



ViT
(frozen)

$\sim 10^{2\text{-}3}$

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SAE
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post-treatment observation
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sparse
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locomotion
body contact
antennae
clustering
grooming
resting
foraging
nest structure
...

interpretation

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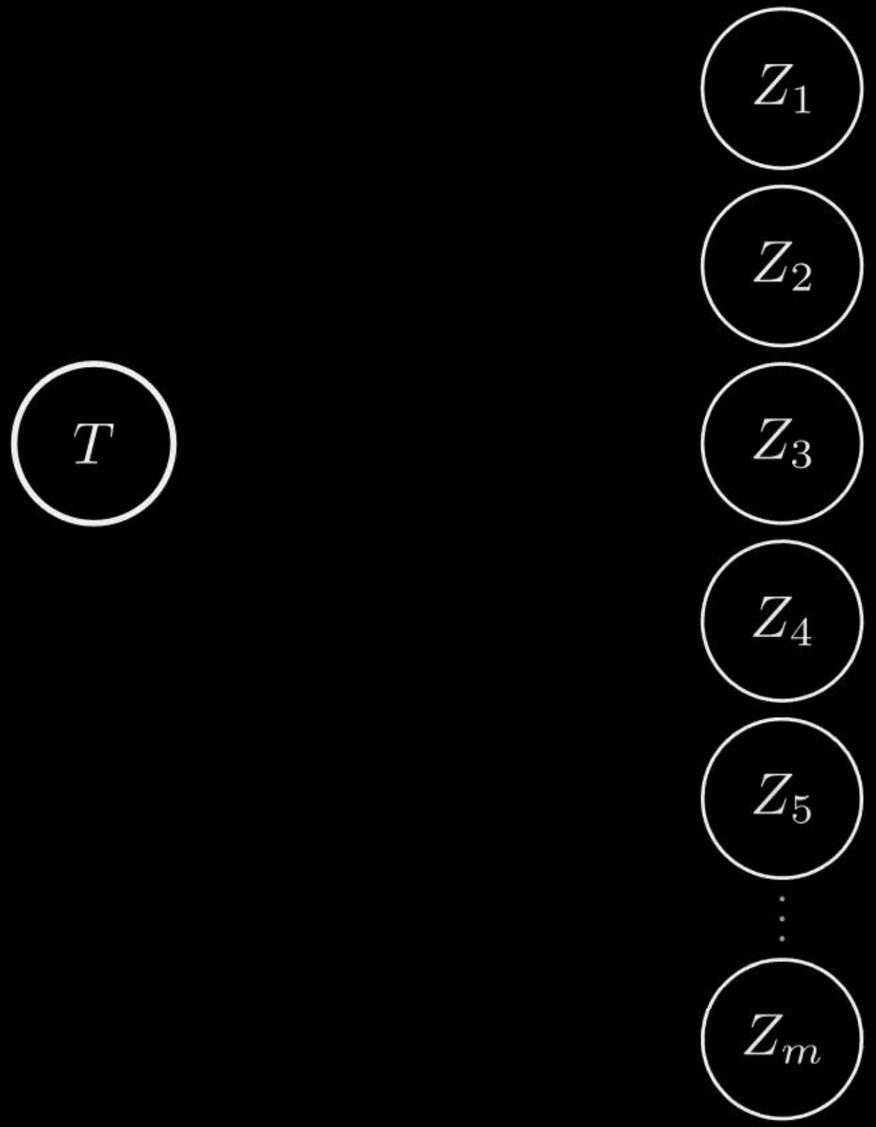


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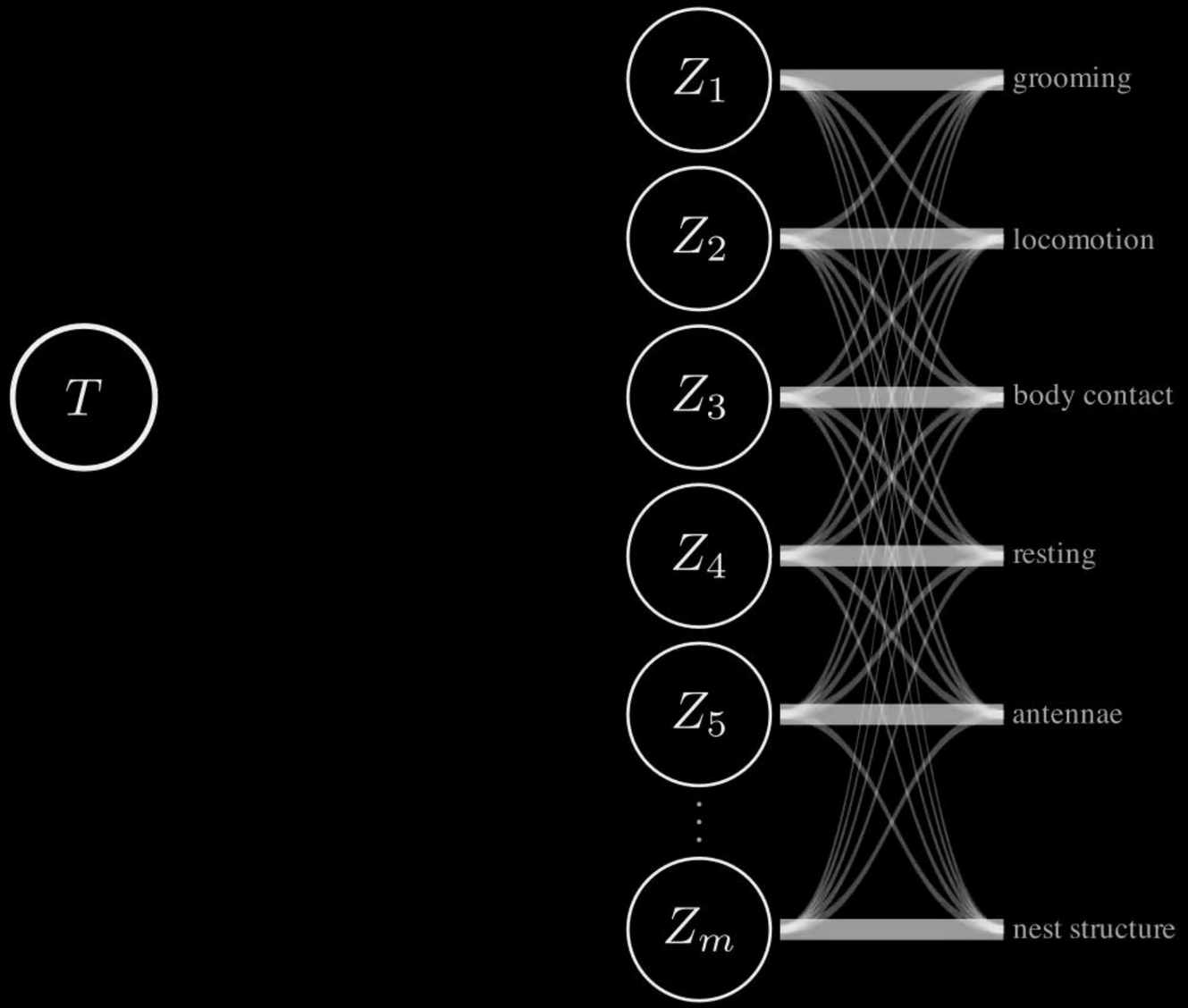


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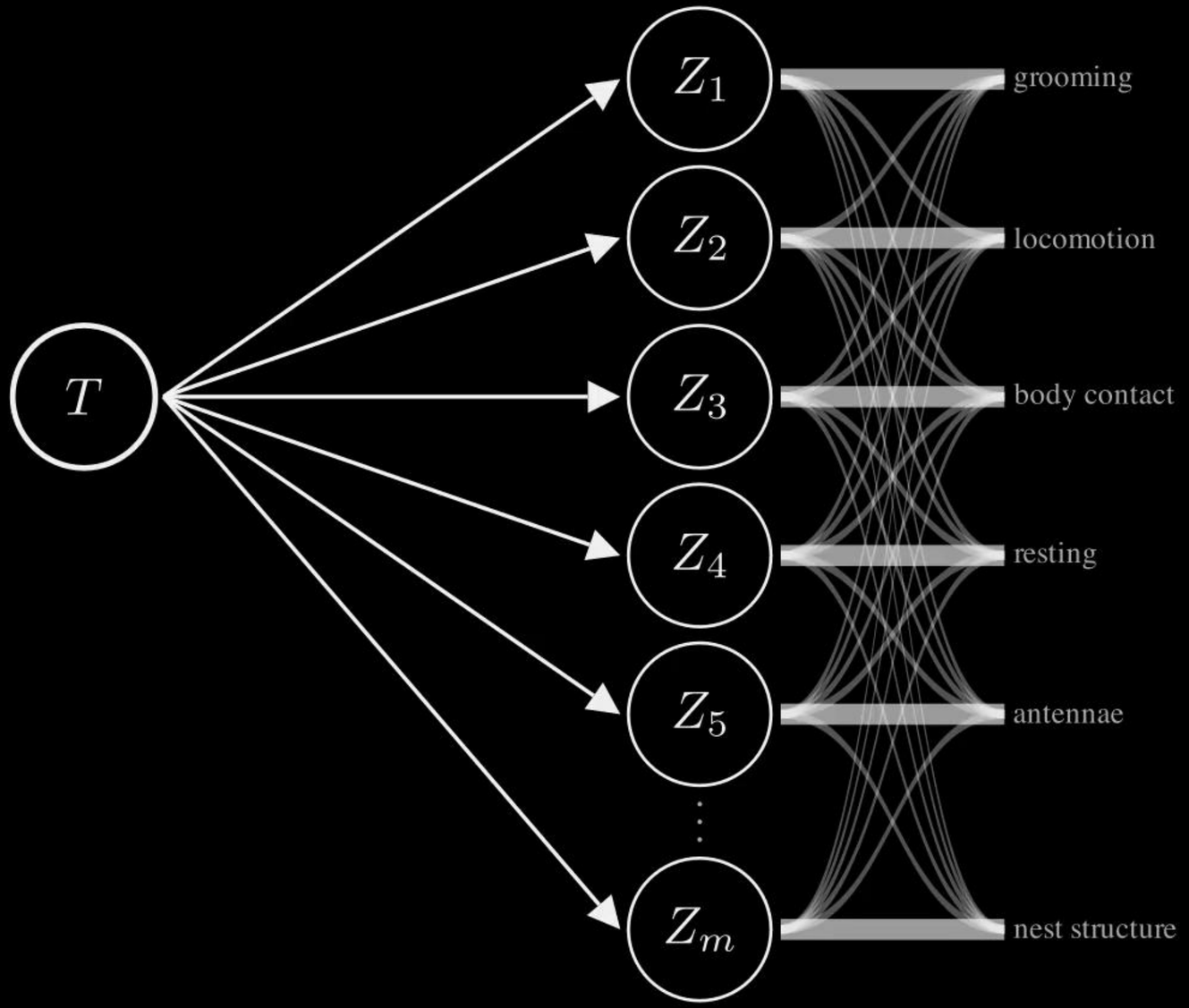
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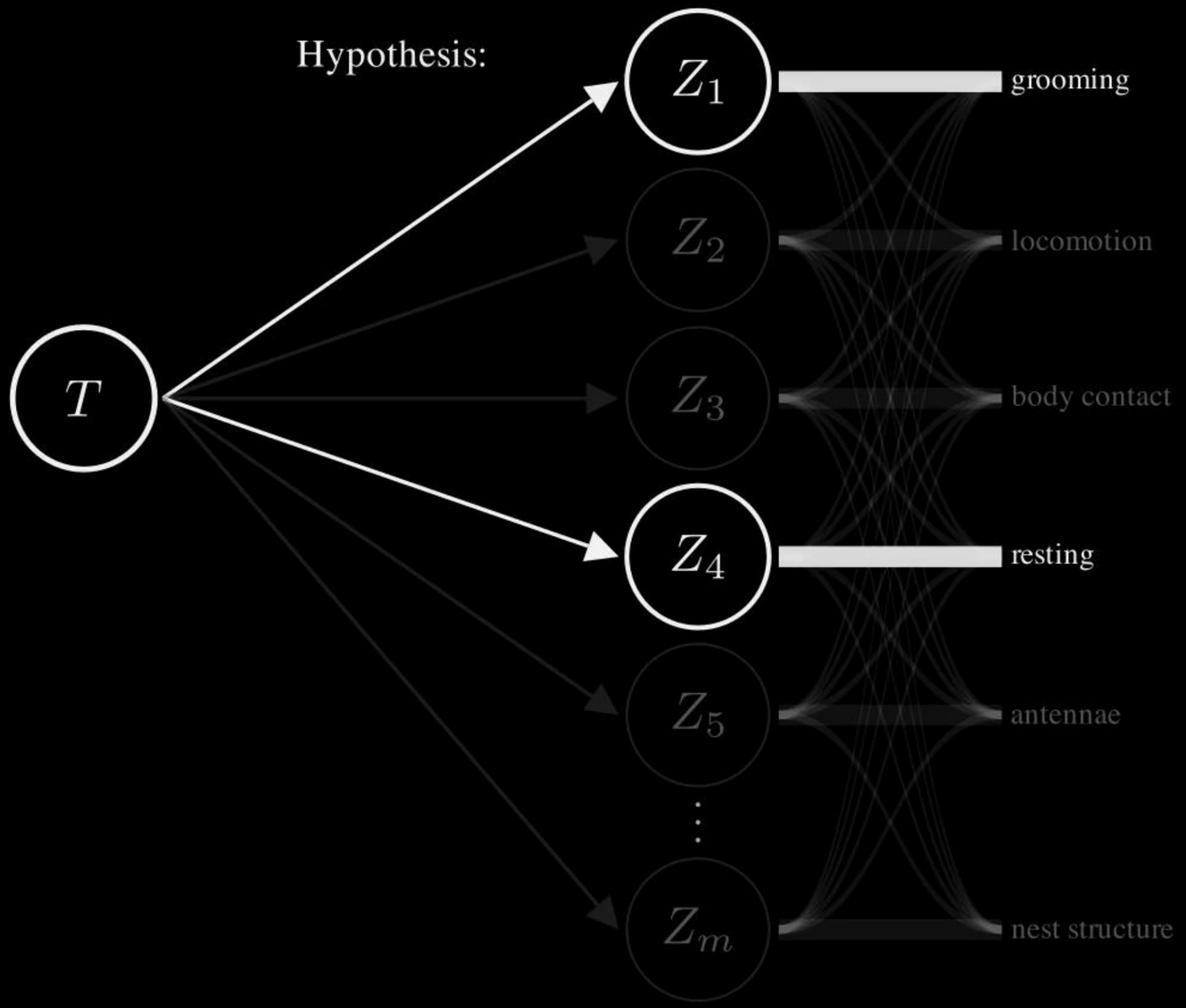
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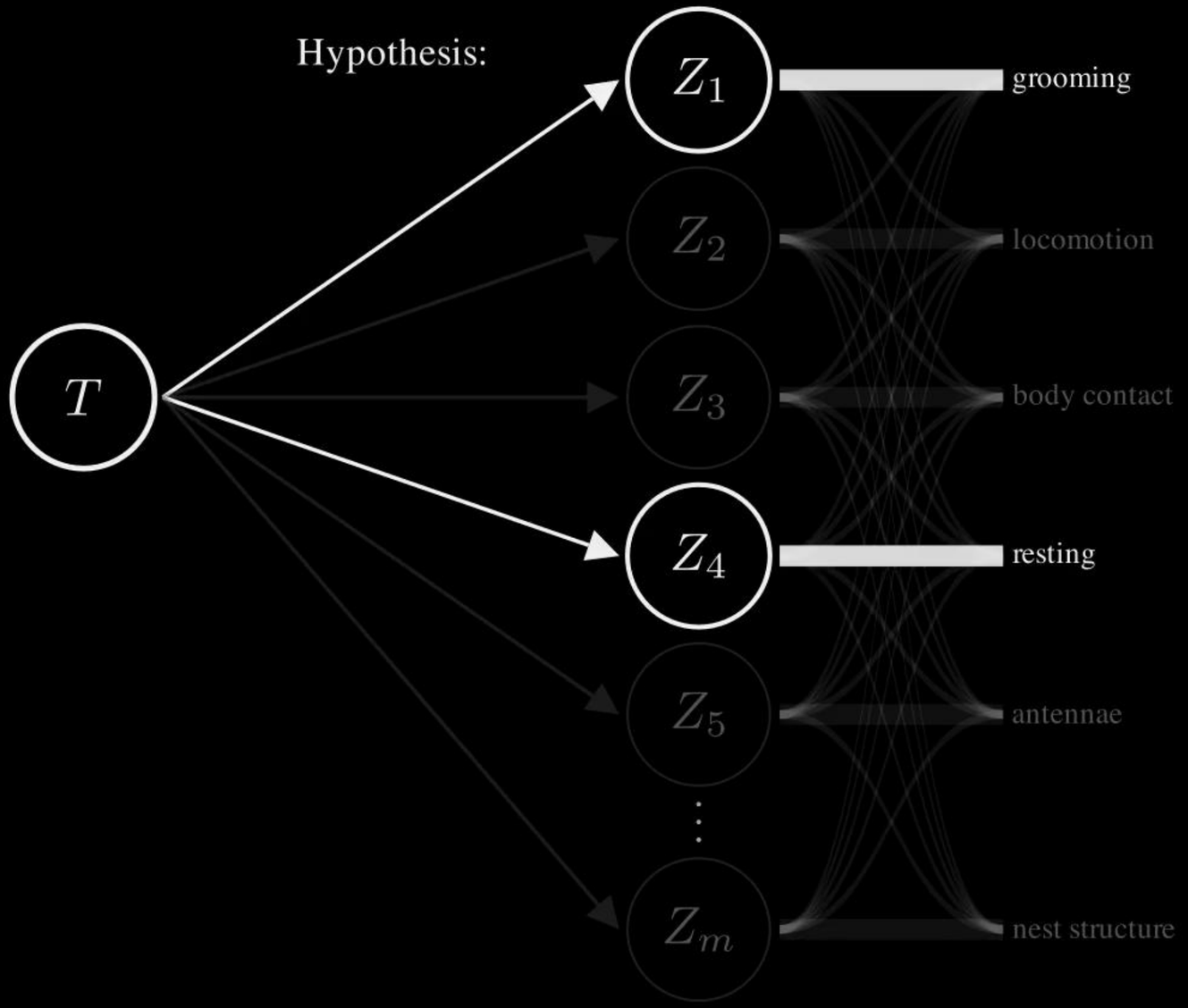
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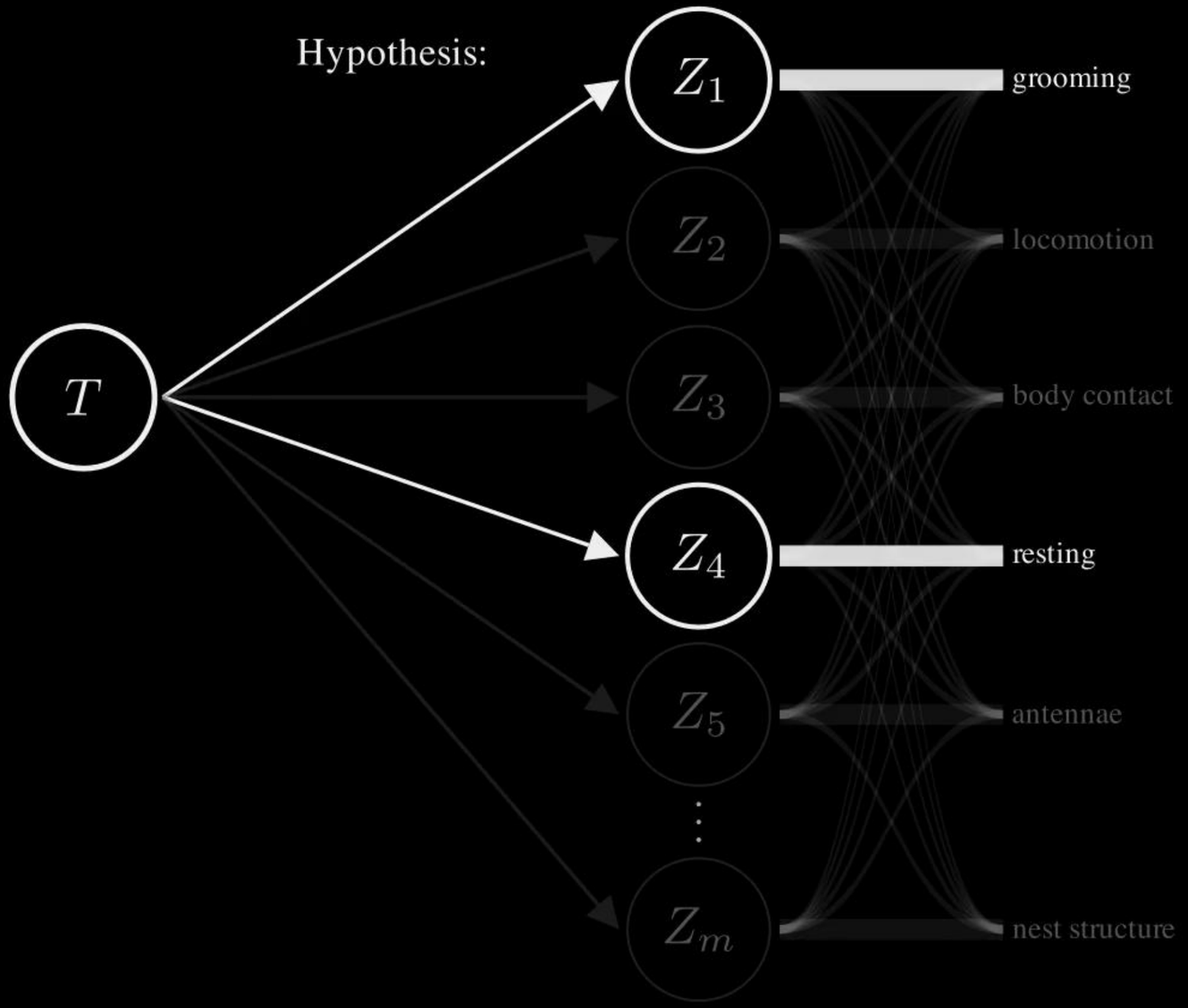
Exploratory Causal Inference



Assumption 1: *Sufficiency*

$$H(Y|X) = H(Y|Z) \approx 0$$

Exploratory Causal Inference



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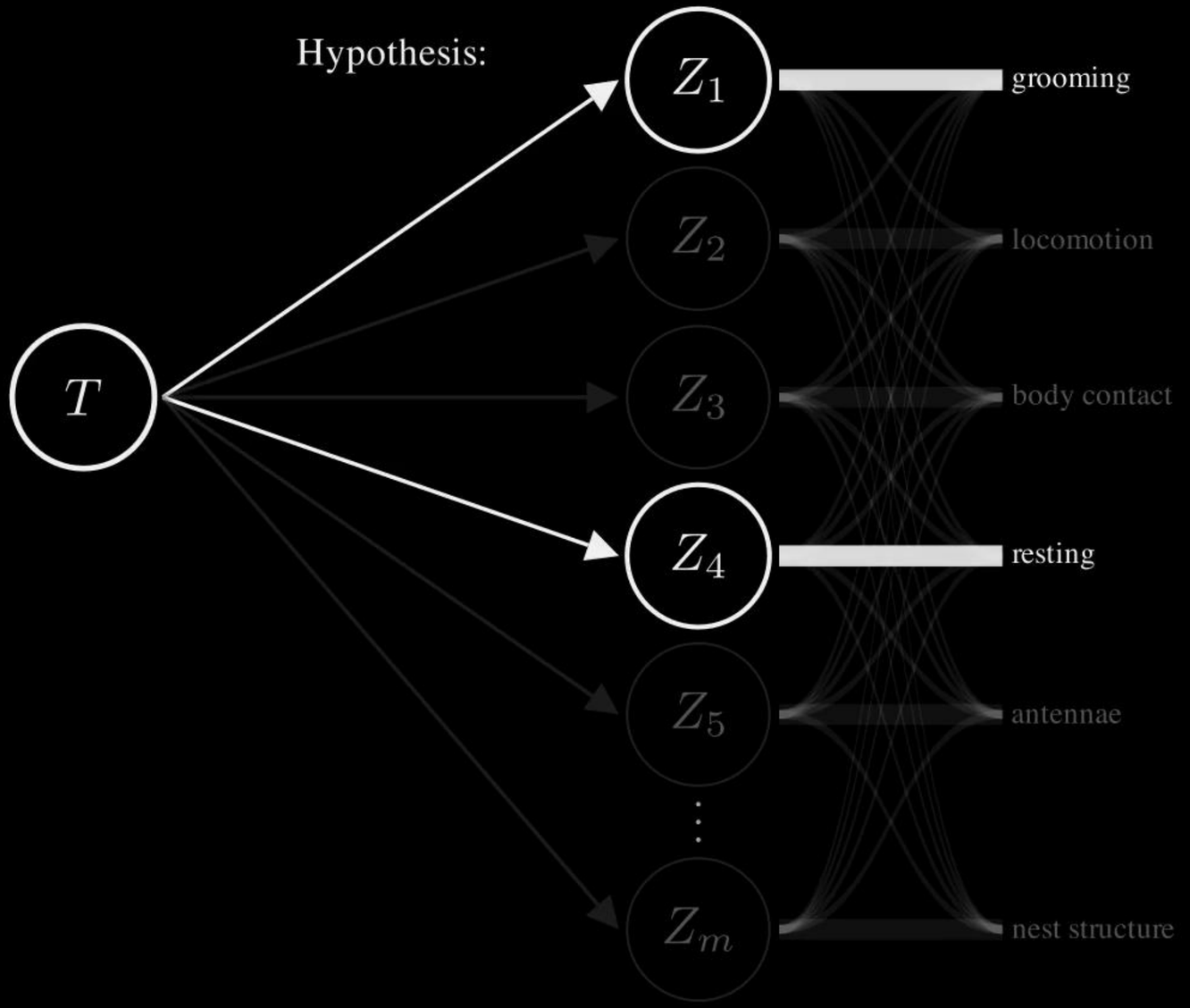
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Assumption 2: *Principal Alignment*

1 Neuron $\approx$ 1 Concept

Exploratory Causal Inference



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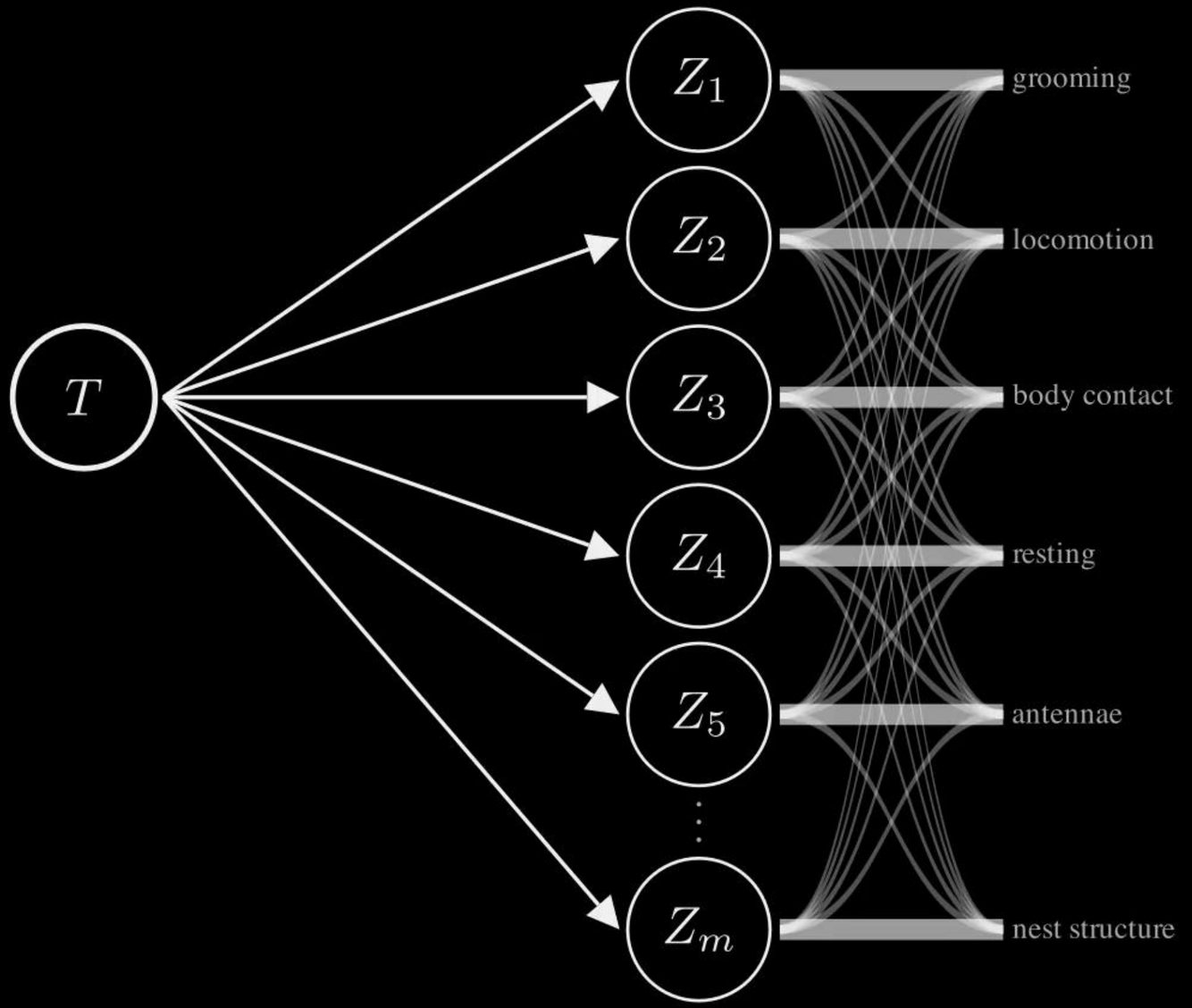
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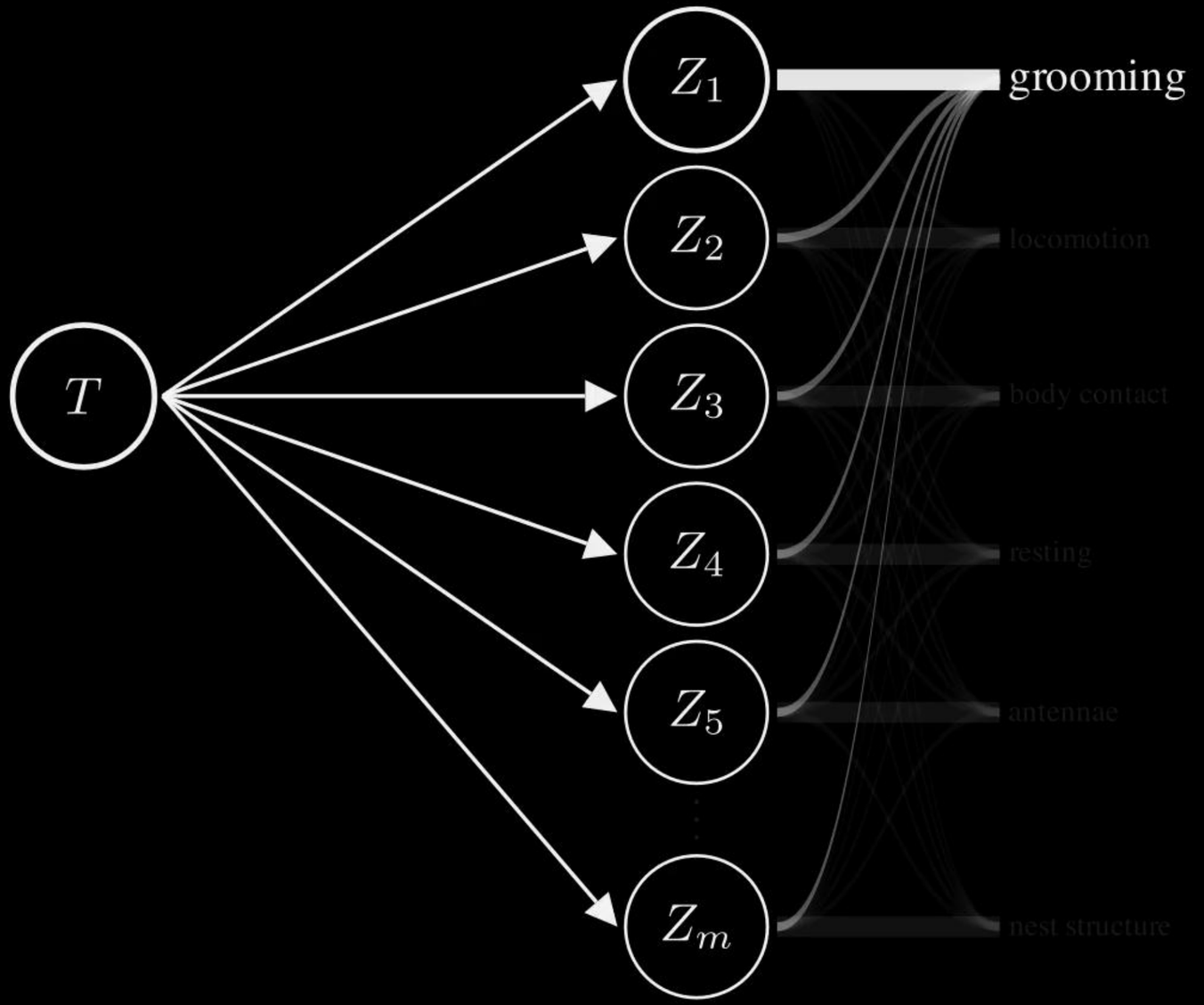


Exploratory Causal Inference Paradox

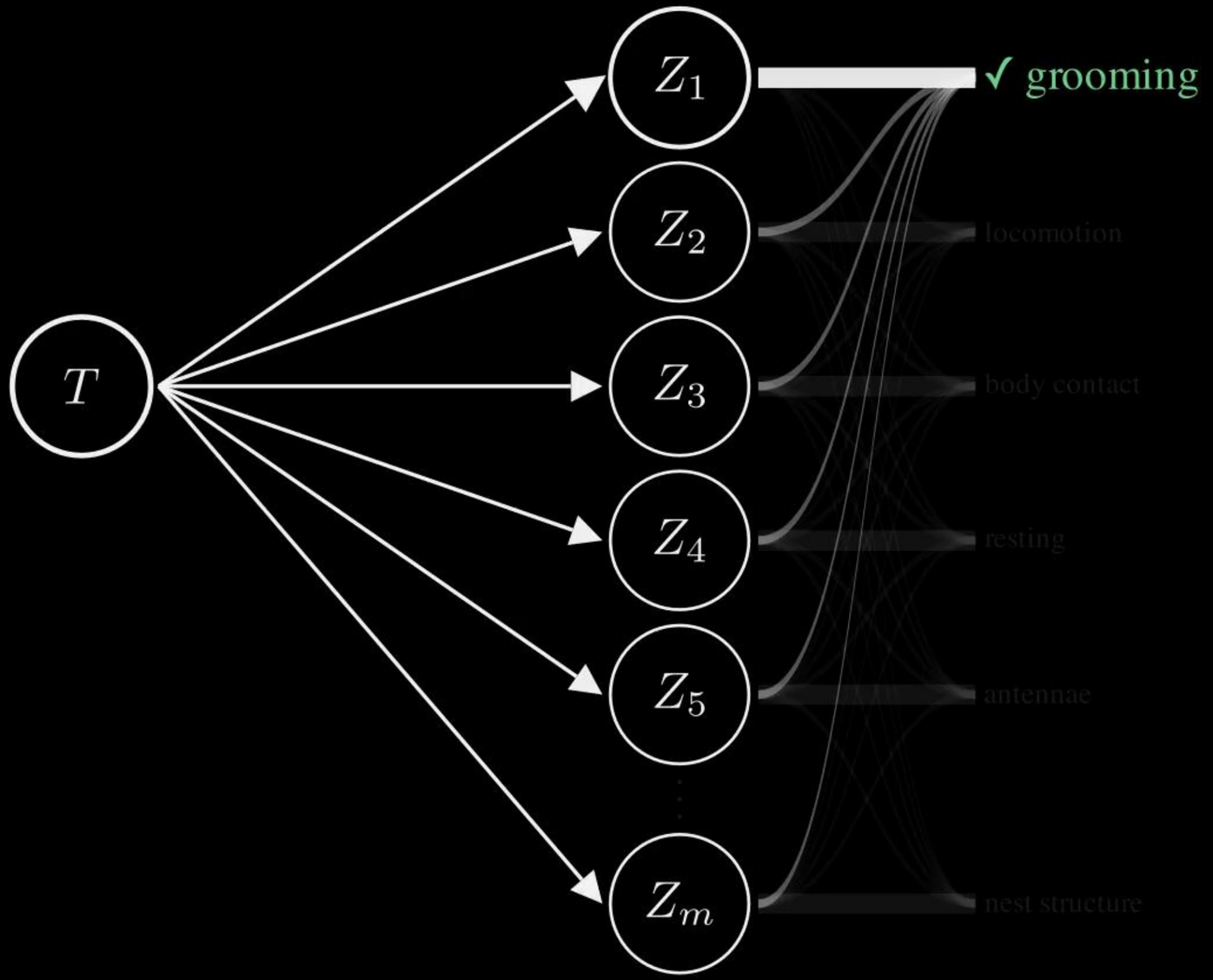
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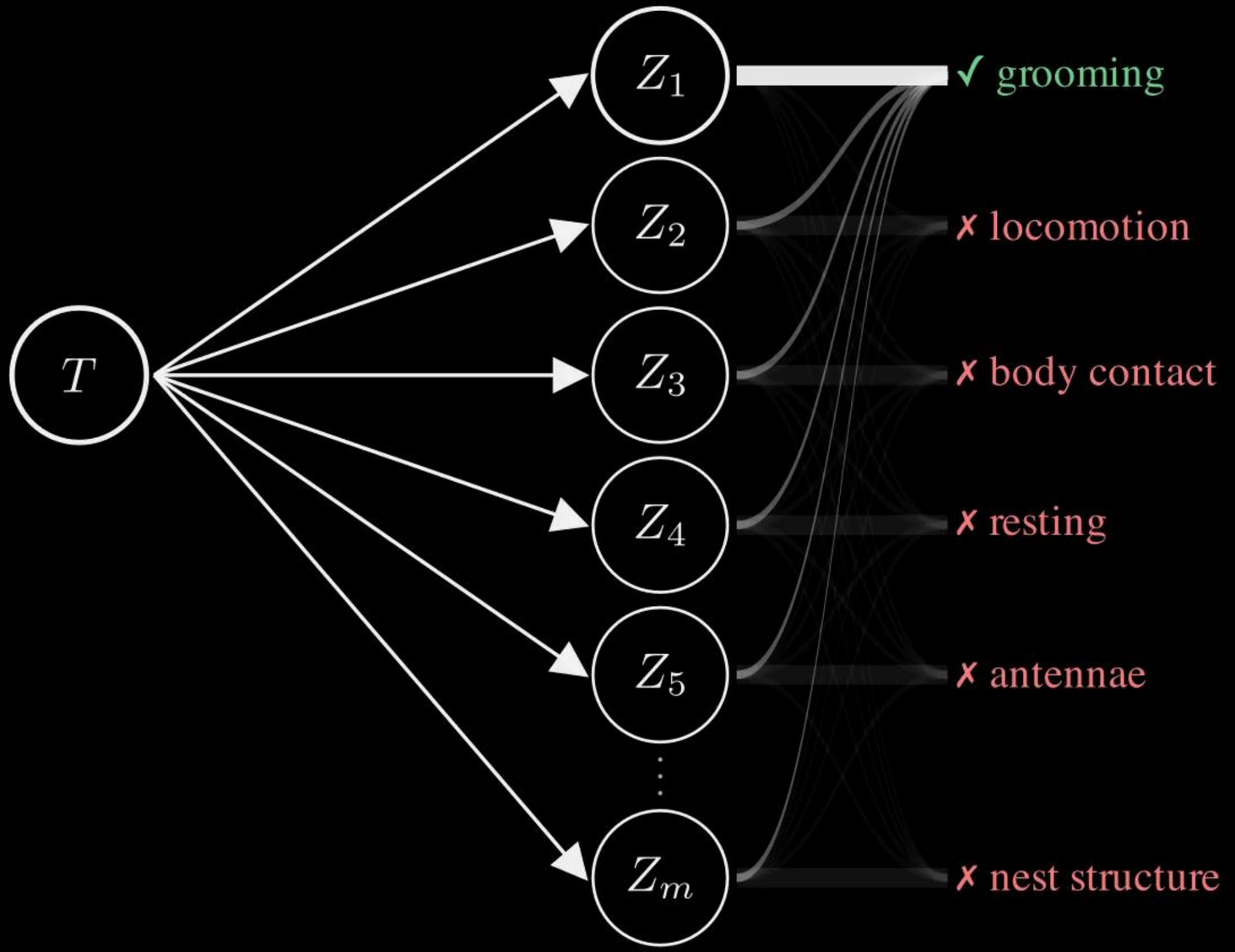
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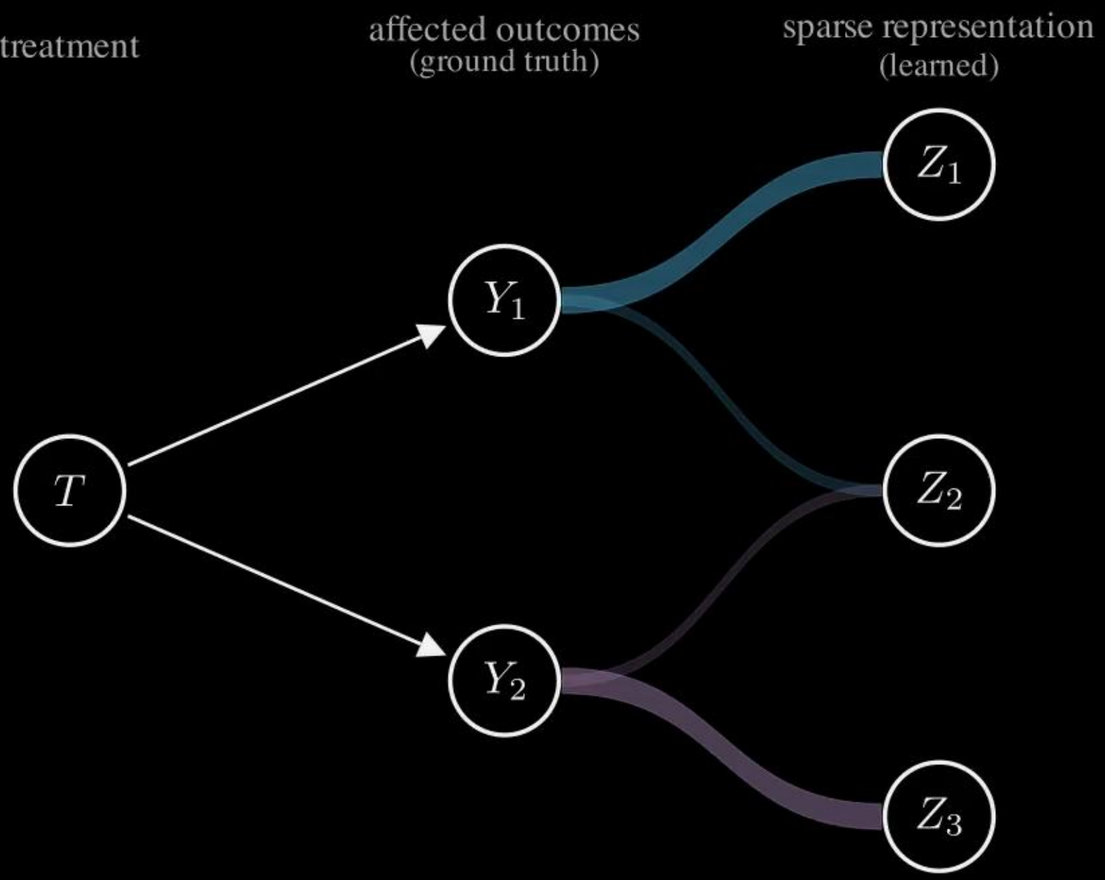


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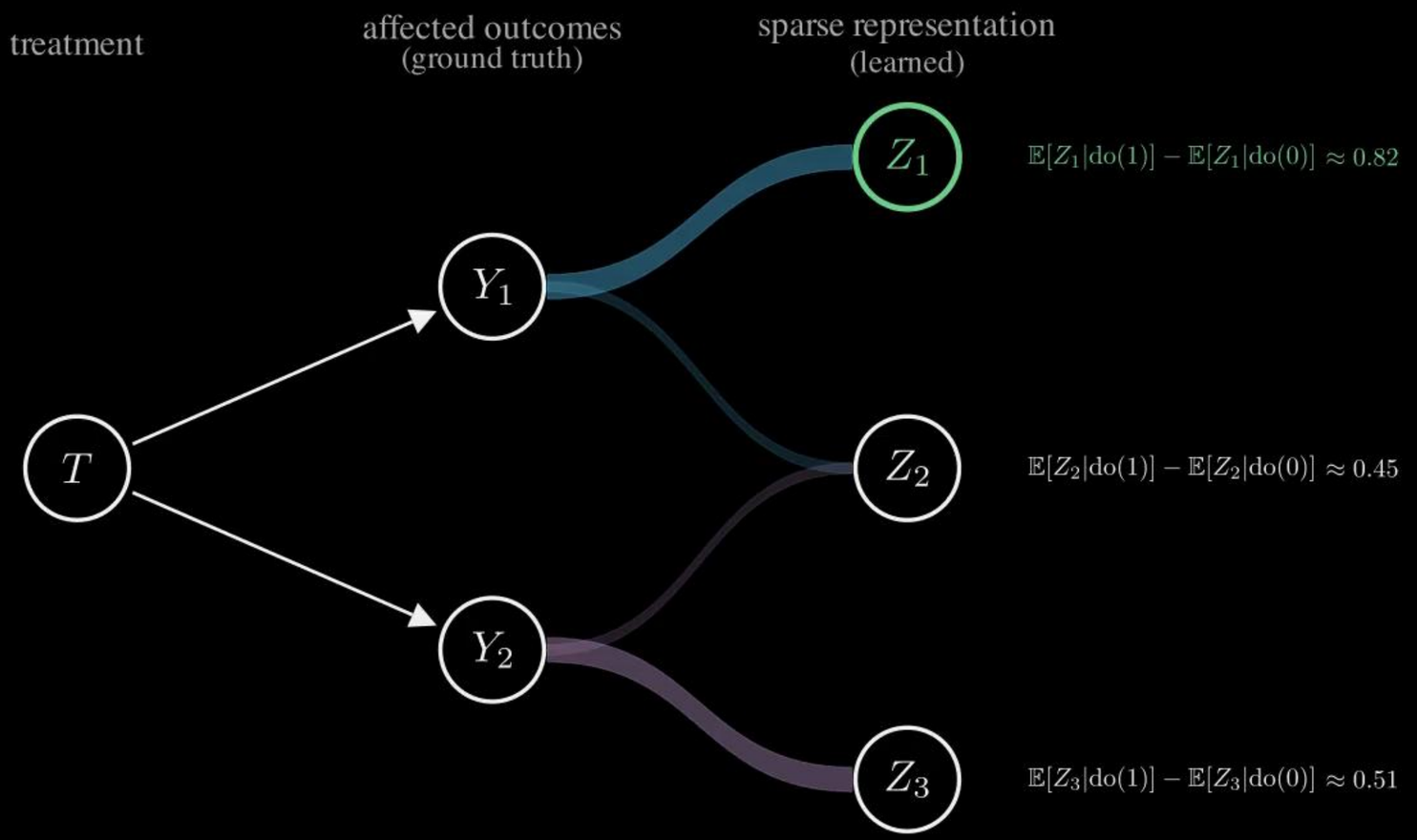


Mitigation: *Neural Effect Search*

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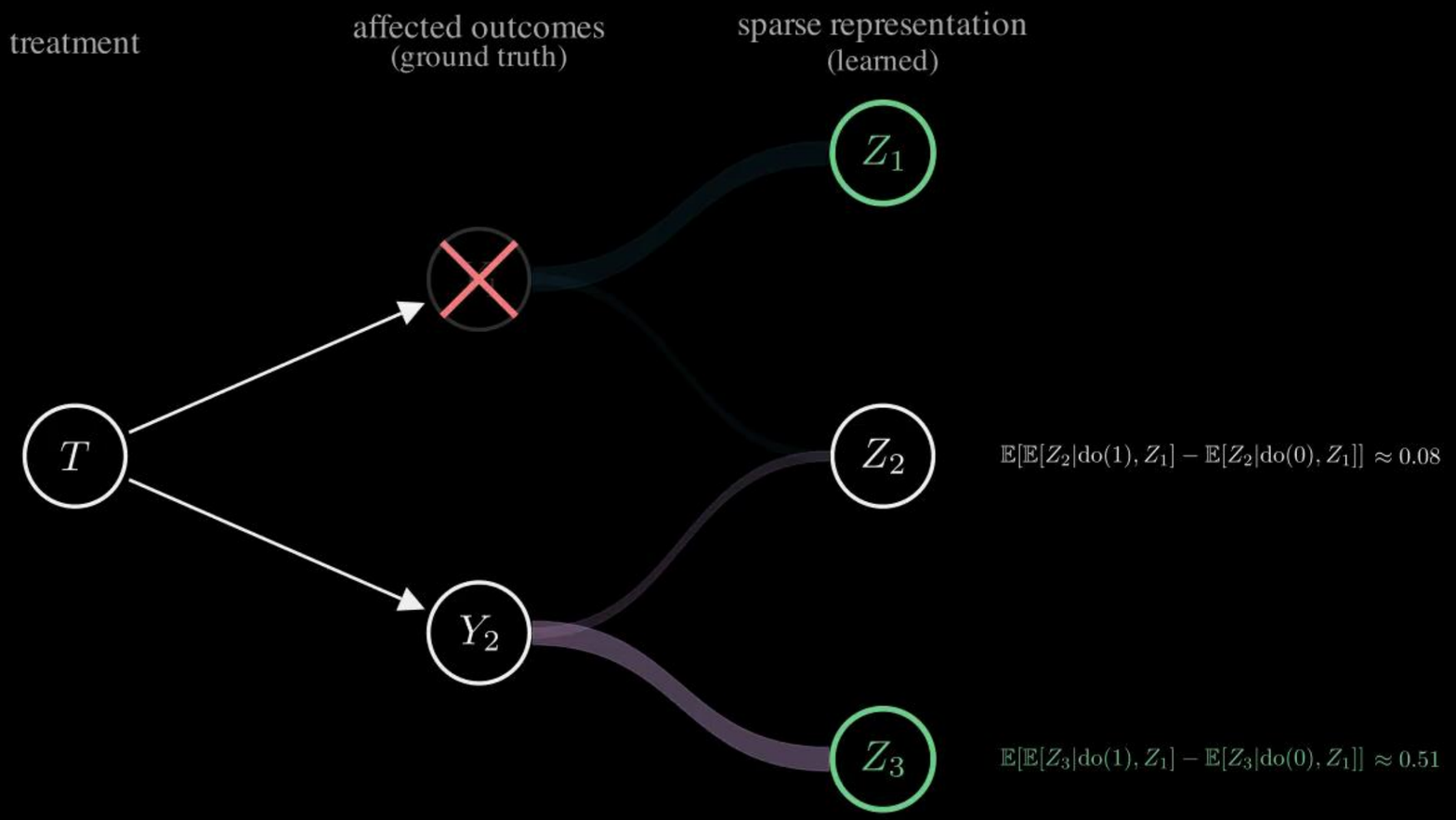


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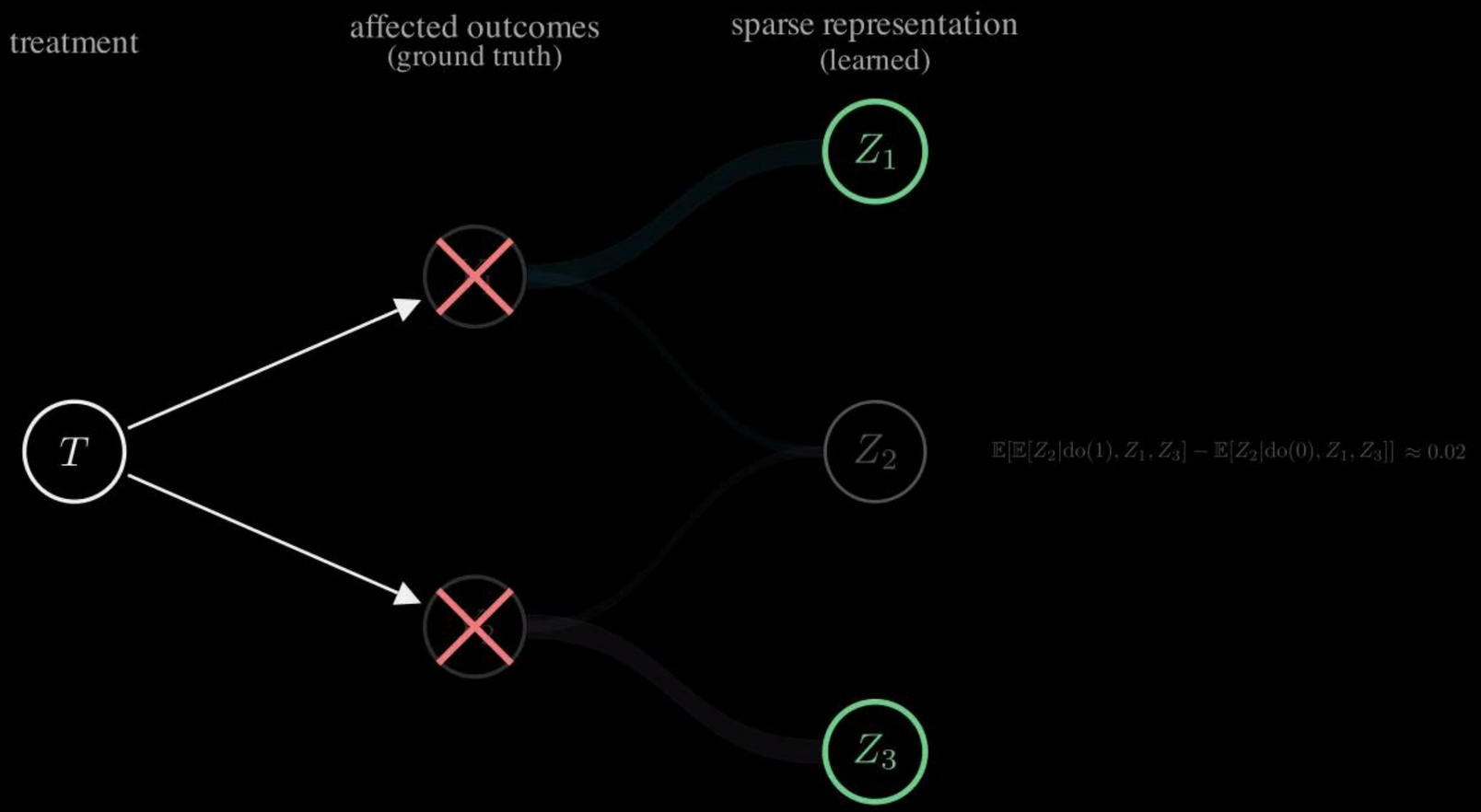
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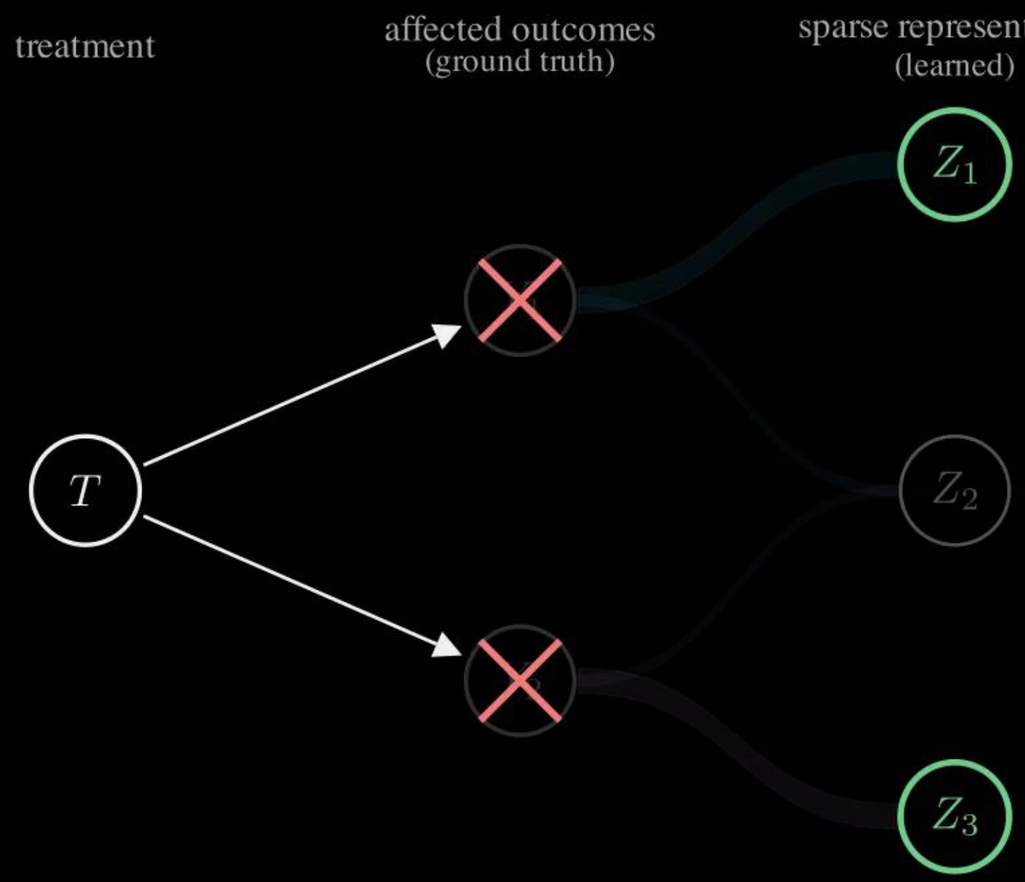
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Step 3: $\mathrm{S}_{\mathrm{final}} = \{Z_1, Z_3\}$

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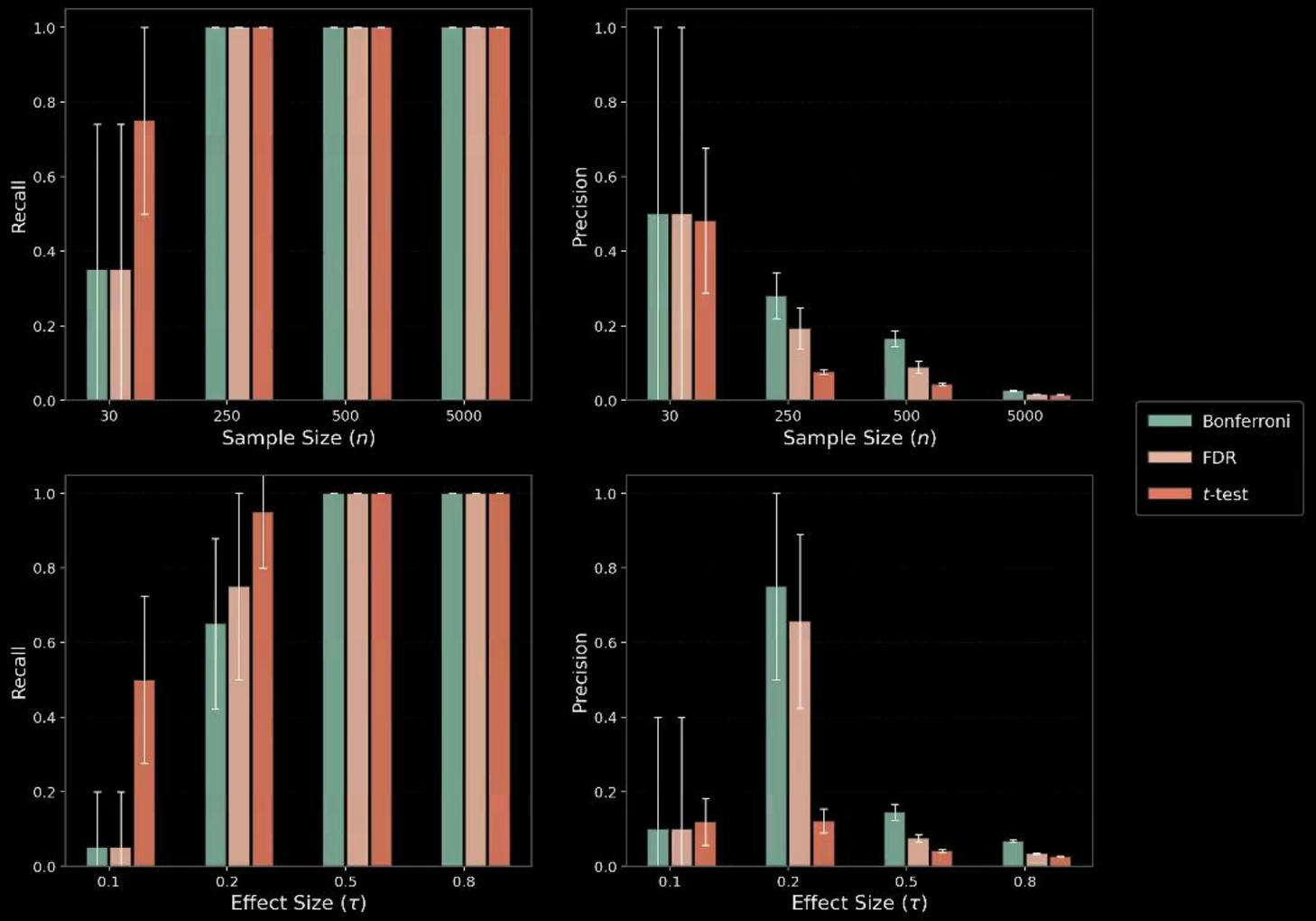
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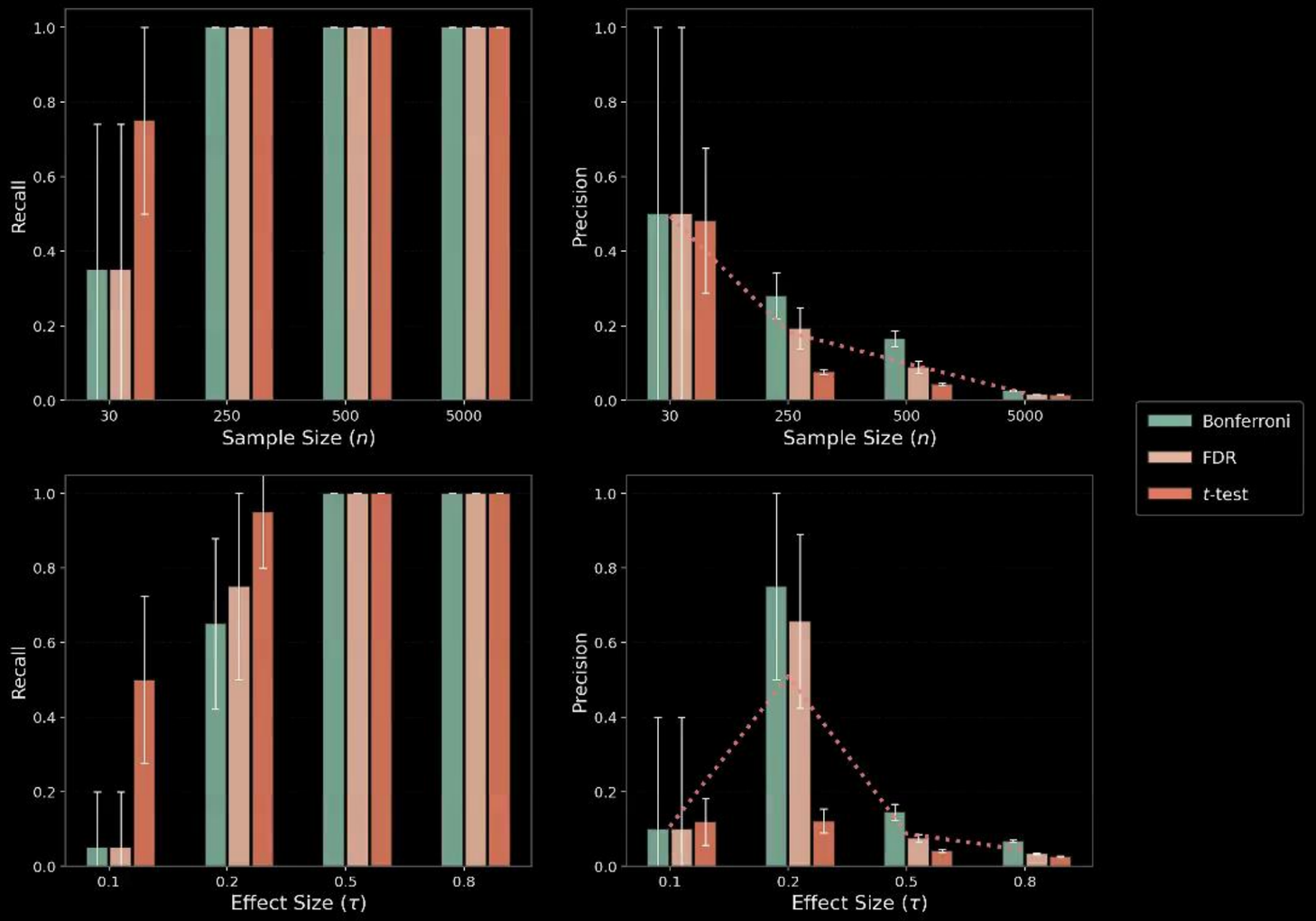
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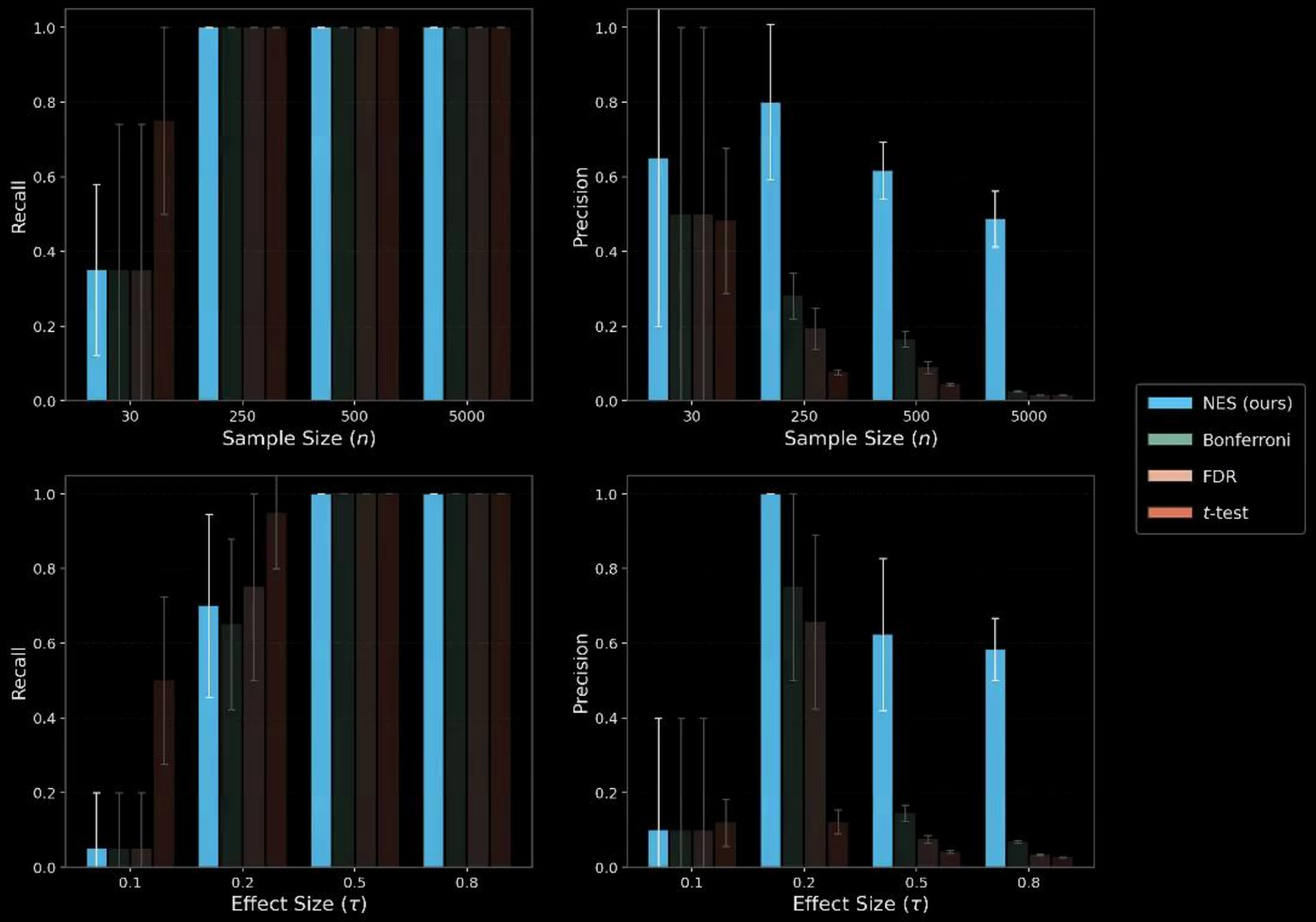


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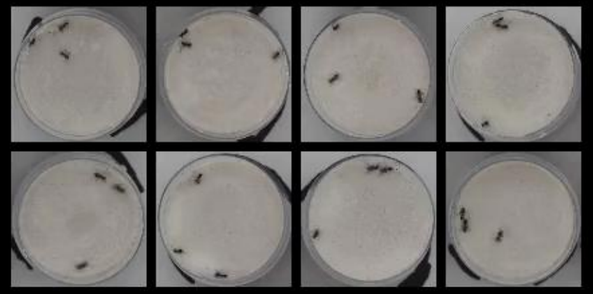
NES (ours): ~~*precision collapse*~~

Results: *Social Immunity (ISTAnt)*

2/4608 significant neural effects

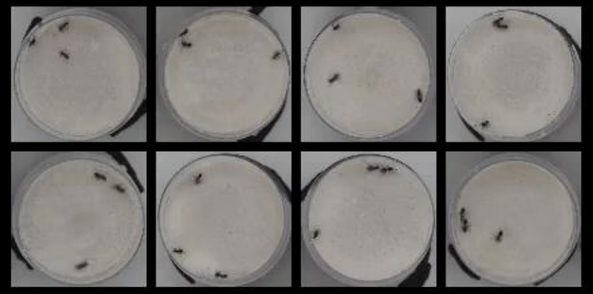
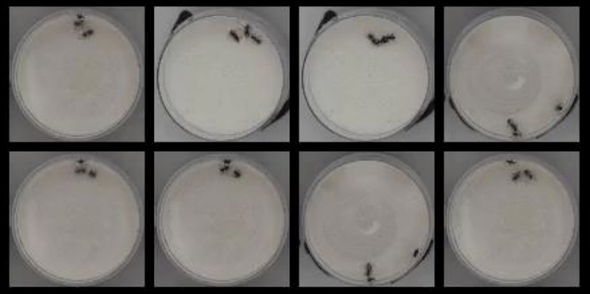
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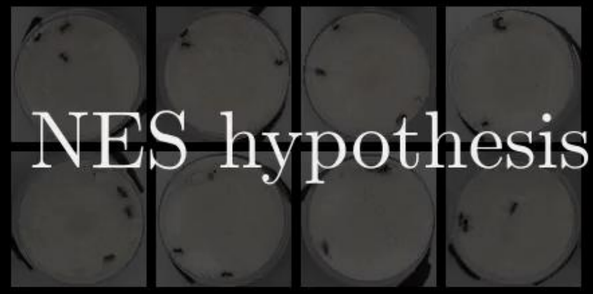
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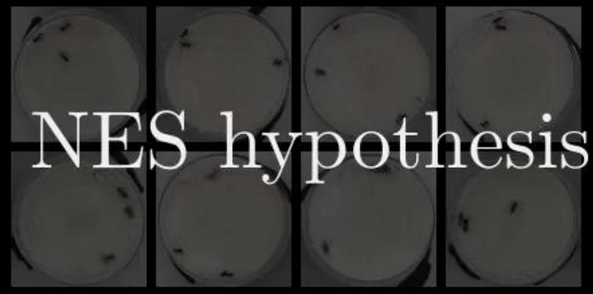
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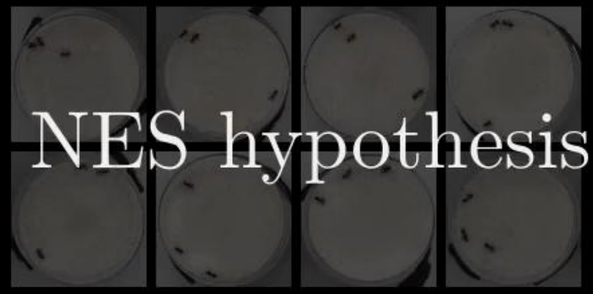
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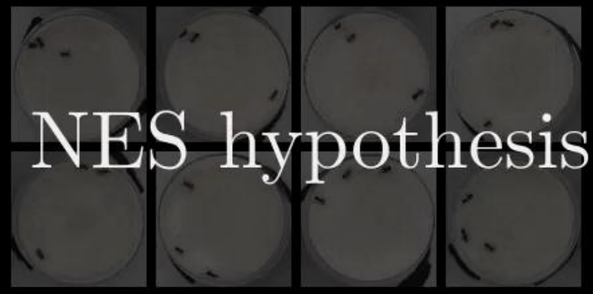
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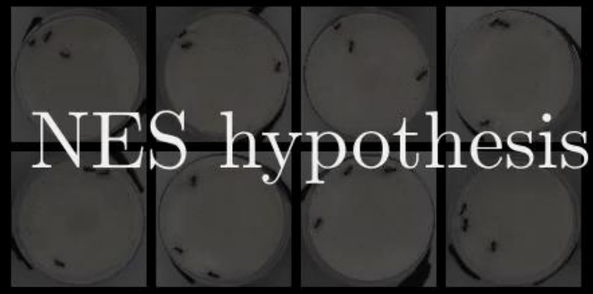
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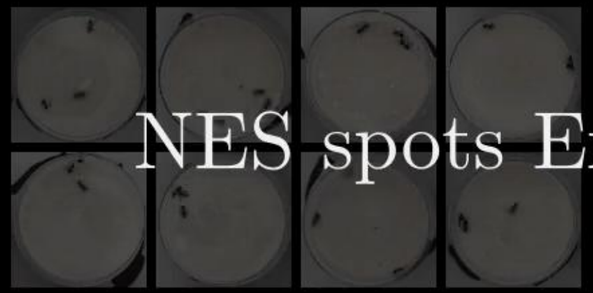
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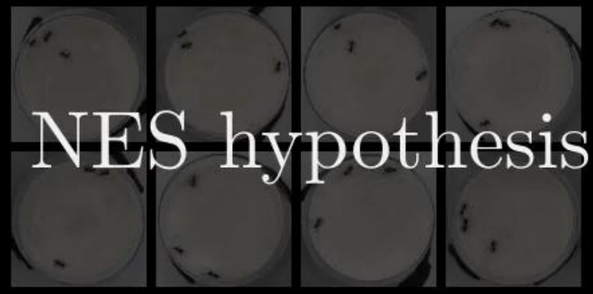
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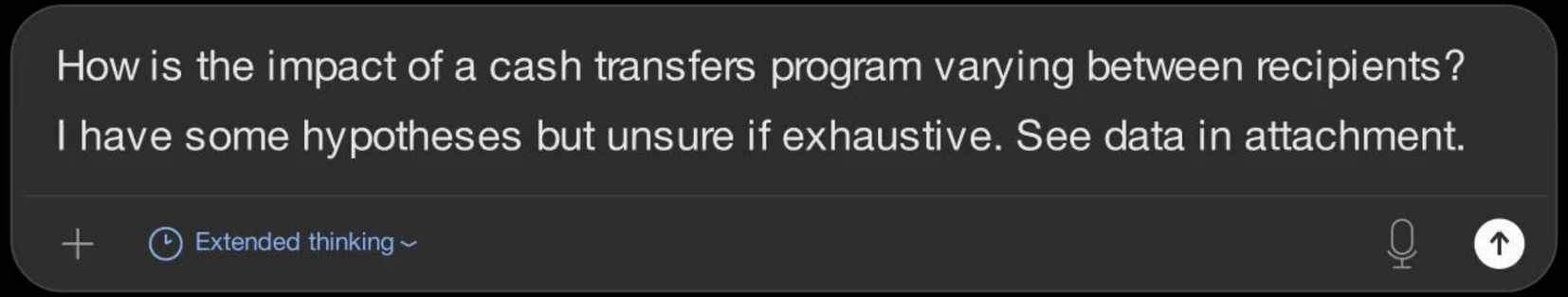
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pre-treatment observation



treatment



affected outcome

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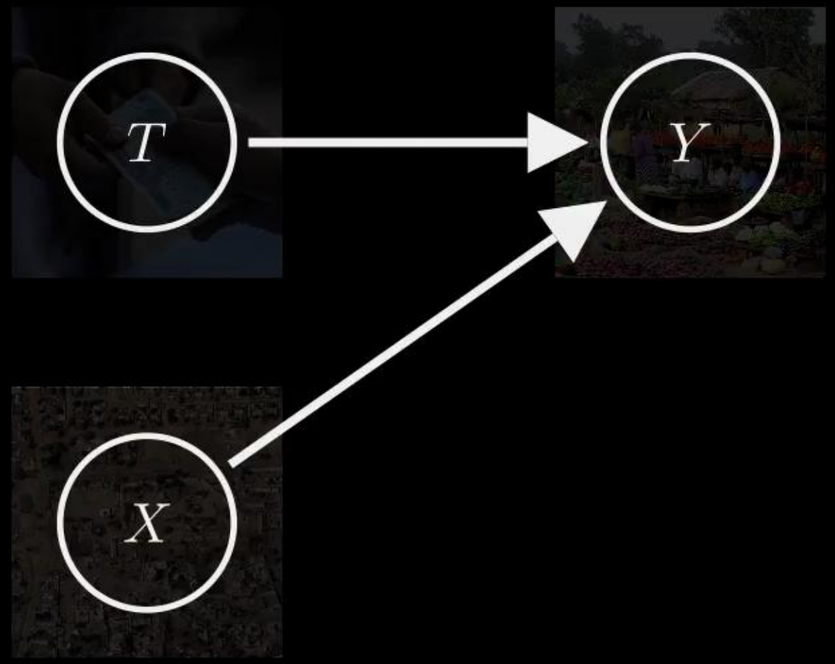


treatment



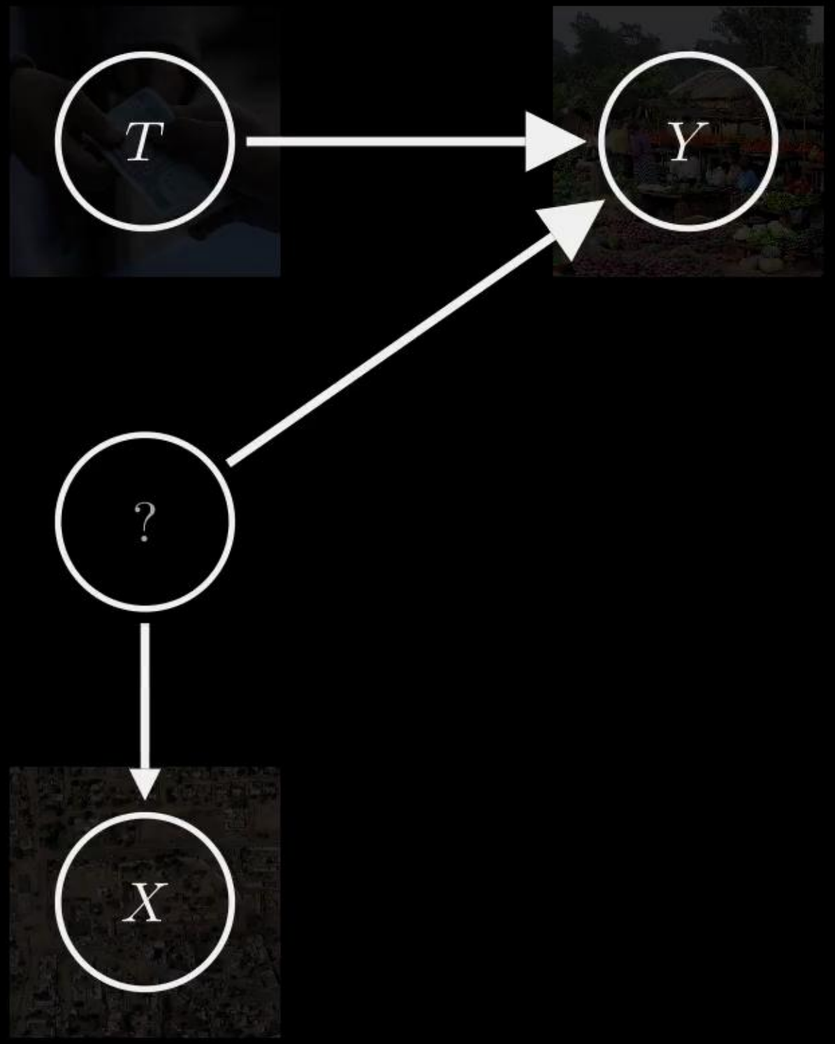
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Exploratory Causal Inference

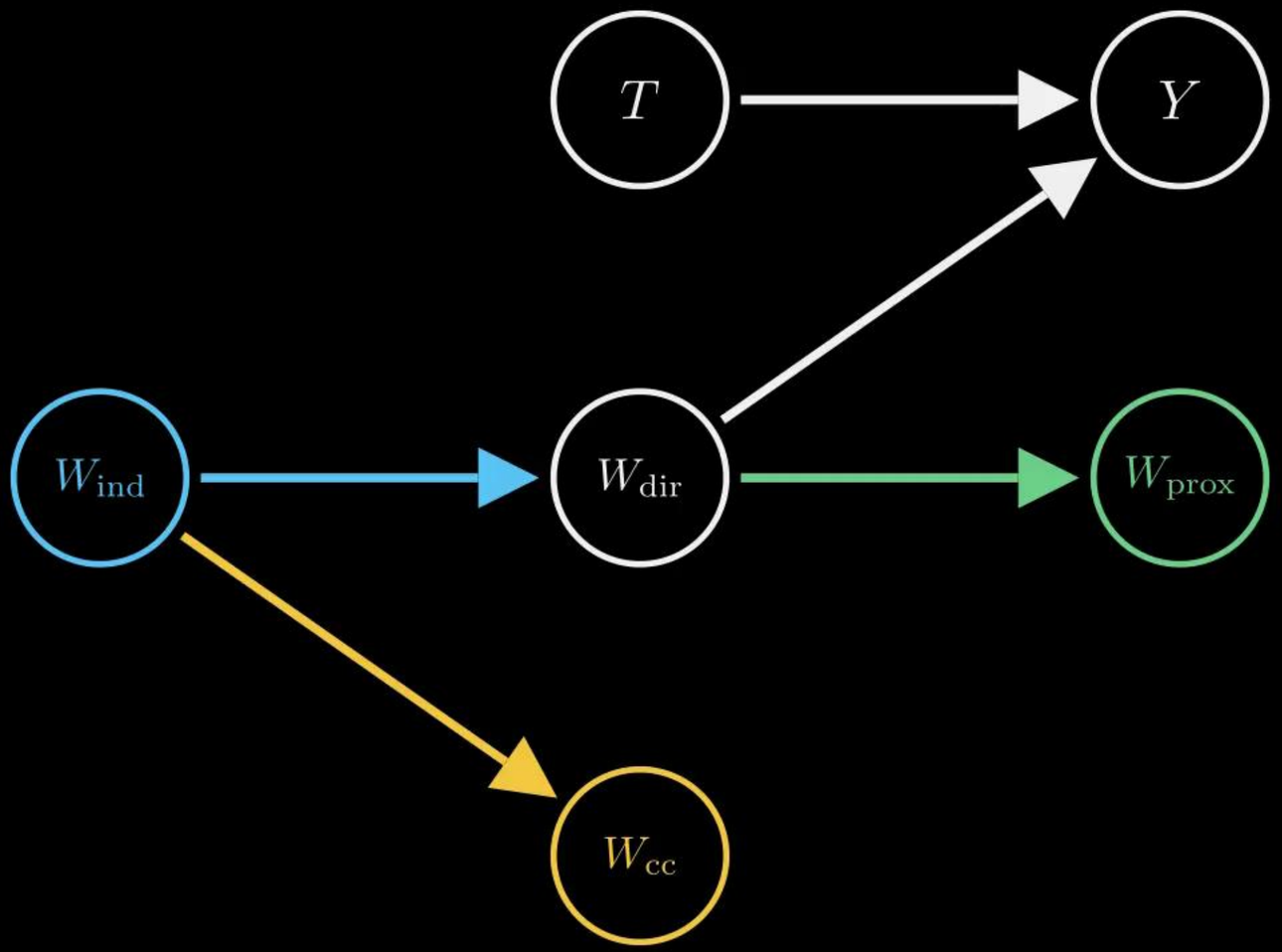


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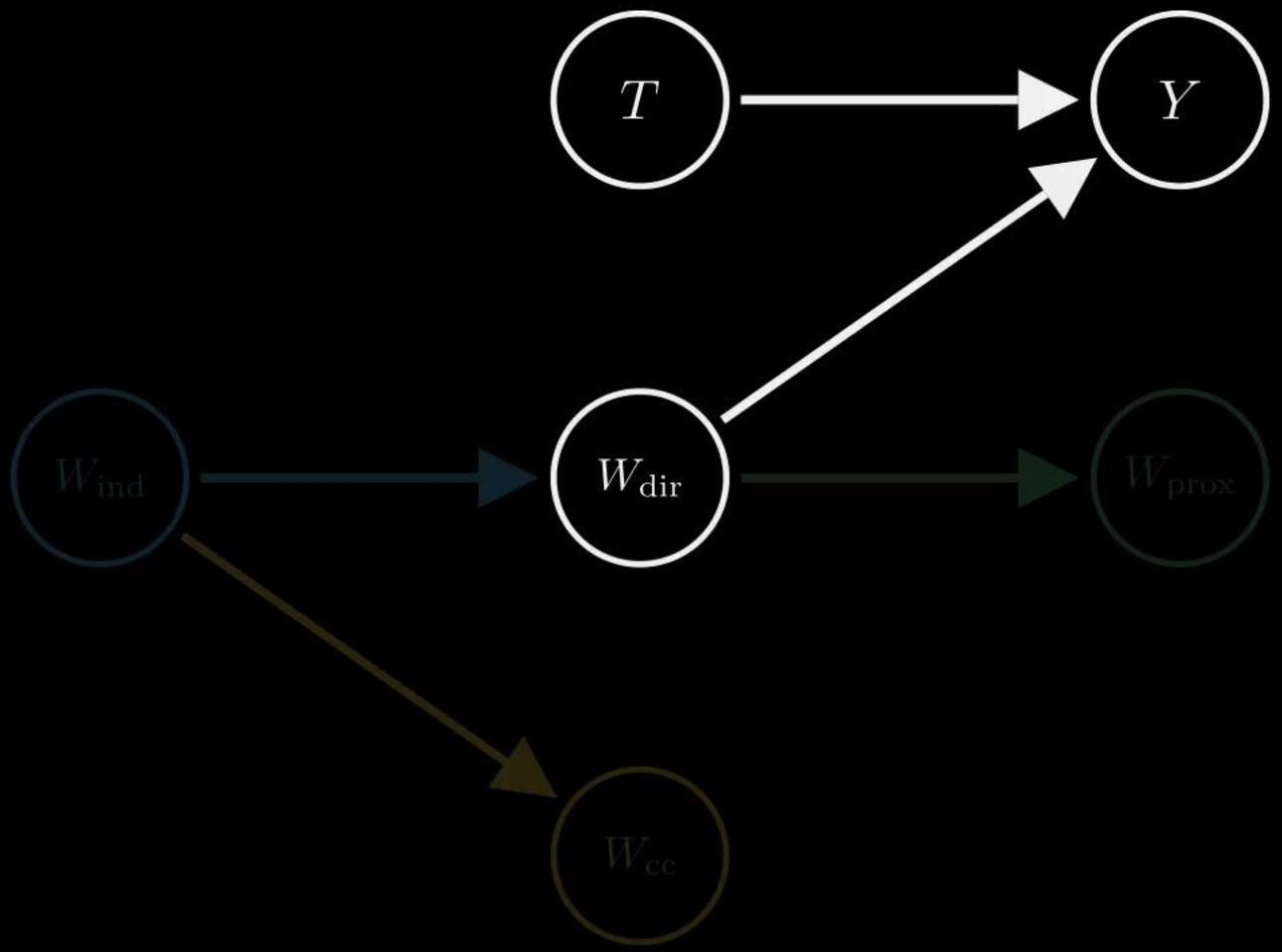
Direct Effect Modifiers Identification



Direct Effect Modifiers Identification



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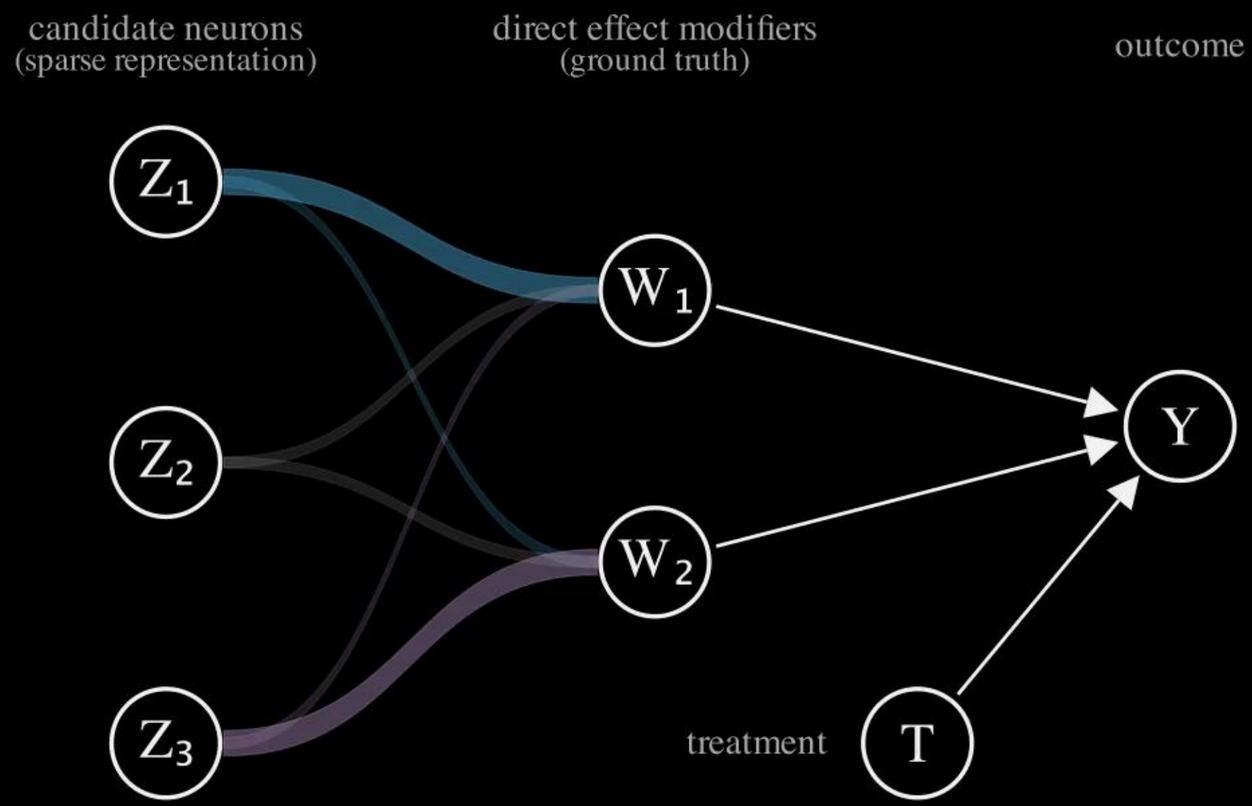
Tool: *Dictionary Learning*

as before

Exploratory Causal Inference Paradox

as before

Neural EXposure Interaction Search



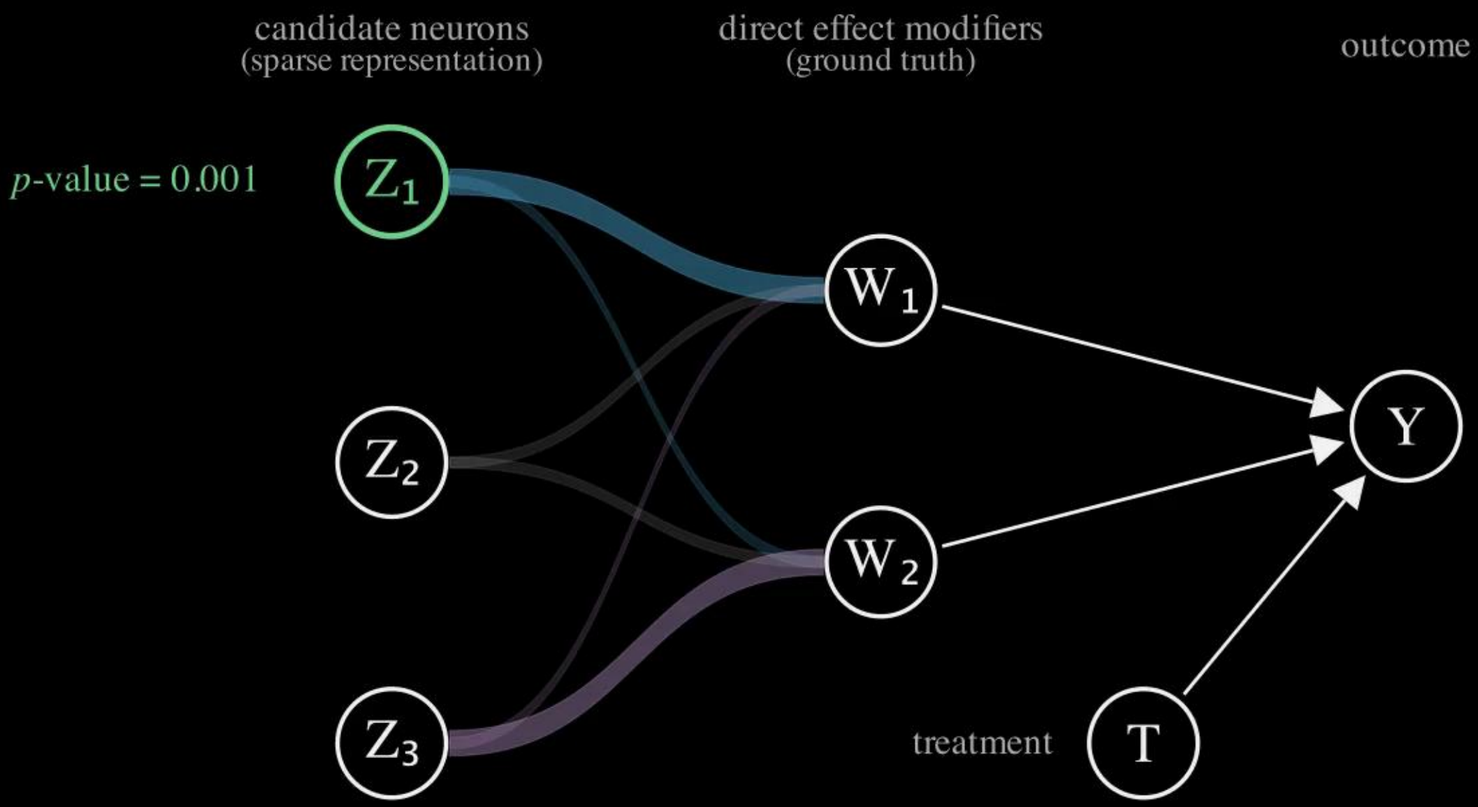
$S = \varnothing$

a. Forward test:

b. Backward test:

Neural EXposure Interaction Search

Step 1



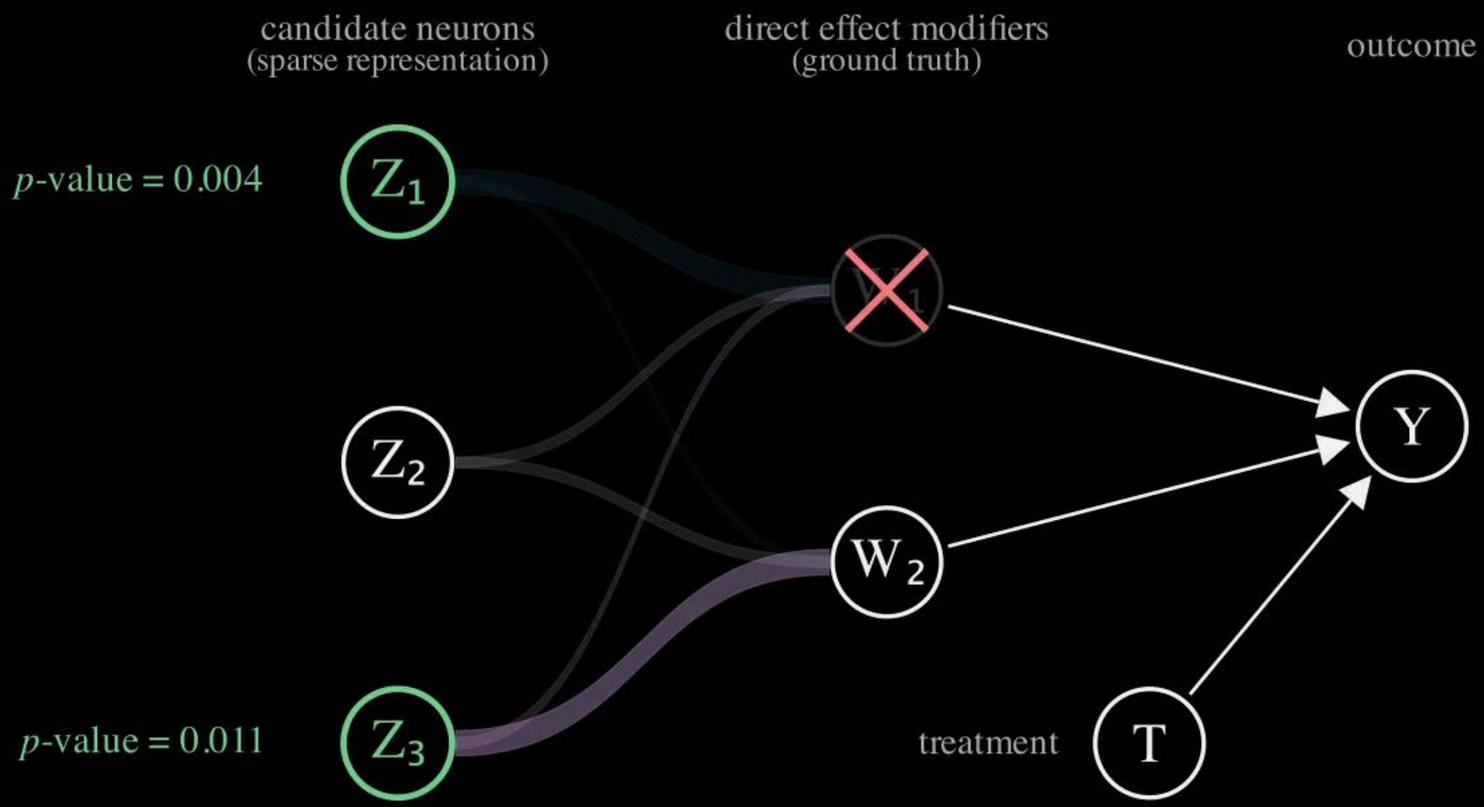
$$S = \{Z_1\}$$

a. Forward test: $H_0 : E[\tau \mid Z_j] = E[\tau] \quad \forall j \in \{1,2,3\}$

b. Backward test: $H_0 : E[\tau \mid Z_1] = E[\tau]$

Neural EXposure Interaction Search

Step 2



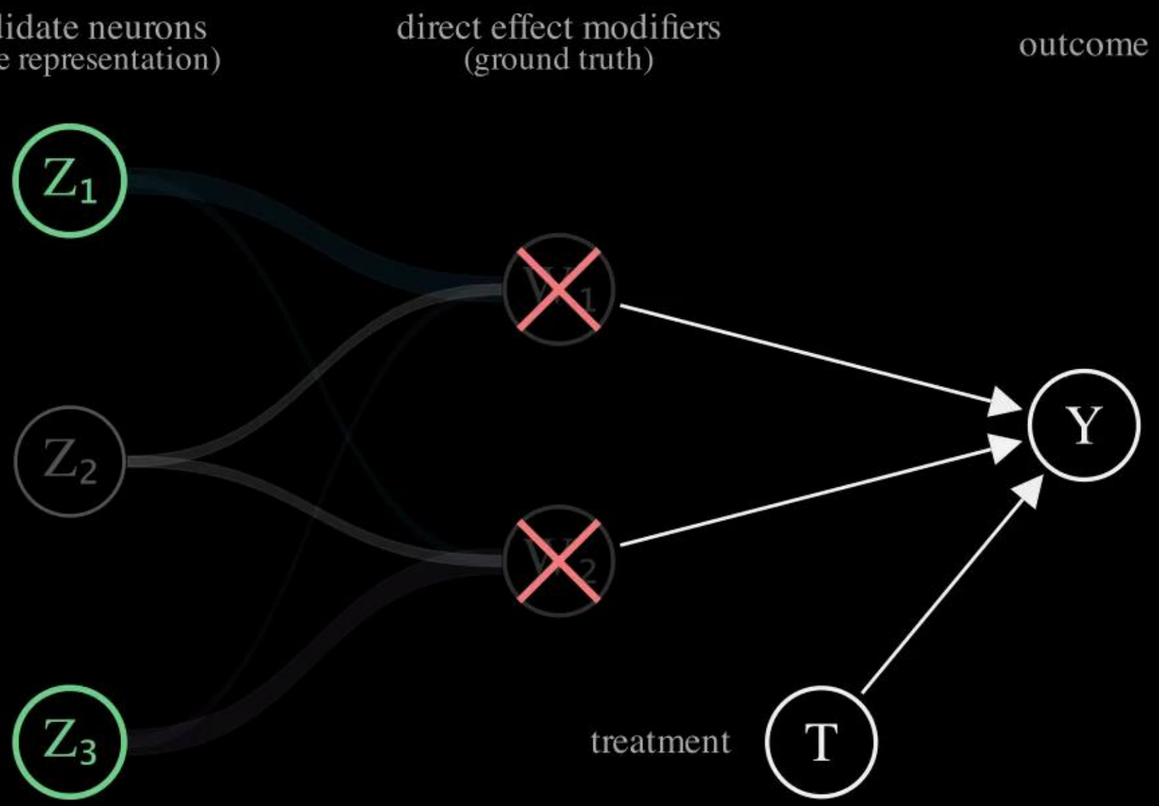
$$S = \{Z_1, Z_3\}$$

a. Forward test: $H_0: E[\tau \mid Z_j, Z_1] = E[\tau \mid Z_1] \quad \forall j \in \{2,3\}$

b. Backward test: $H_0: E[\tau \mid Z_s] = E[\tau \mid Z_{s \setminus j}] \quad \forall j \in \{1,3\}$

Neural EXposure Interaction Search

Step 3

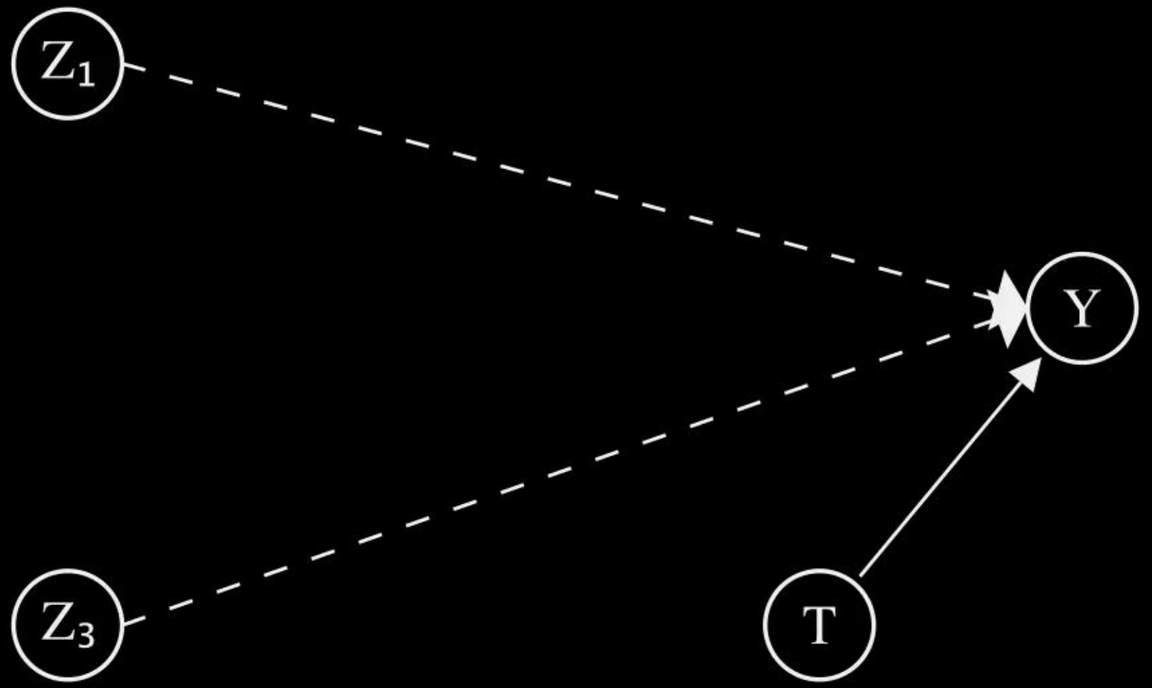


$$S = \{Z_1, Z_3\}$$

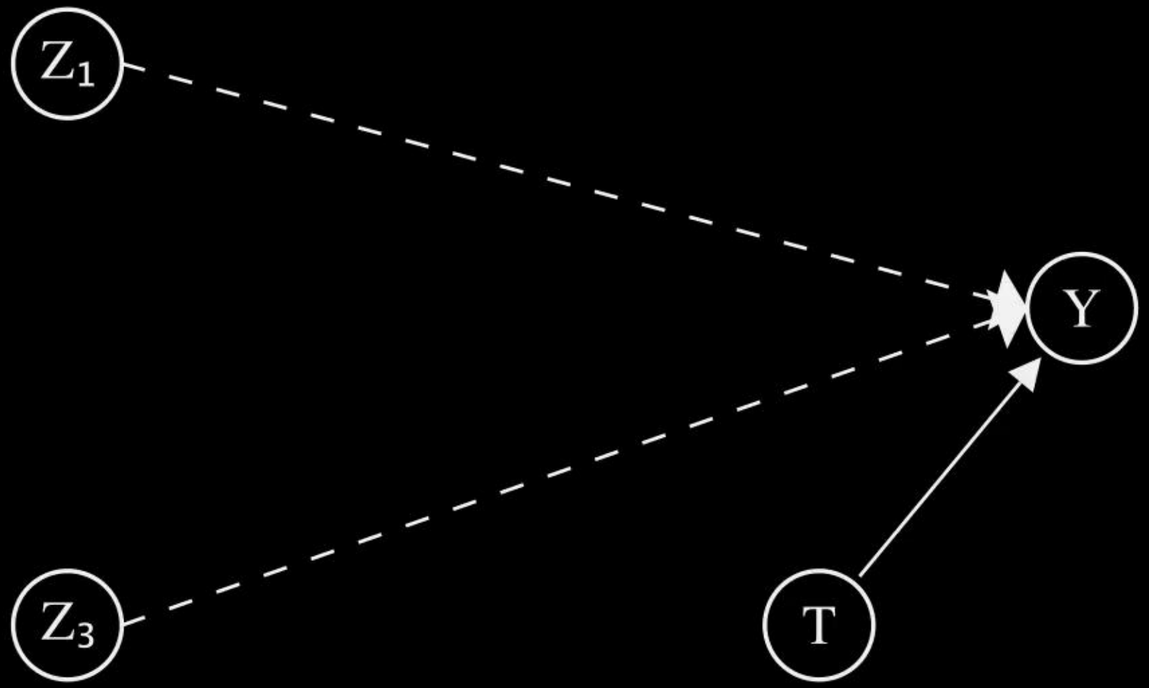
a. Forward test: $H_0: E[\tau \mid Z_1, Z_2, Z_3] = E[\tau \mid Z_1, Z_3]$

b. Backward test:

Neural EXposure Interaction Search



Neural EXposure Interaction Search



Assumption 1: *Sufficiency*

+

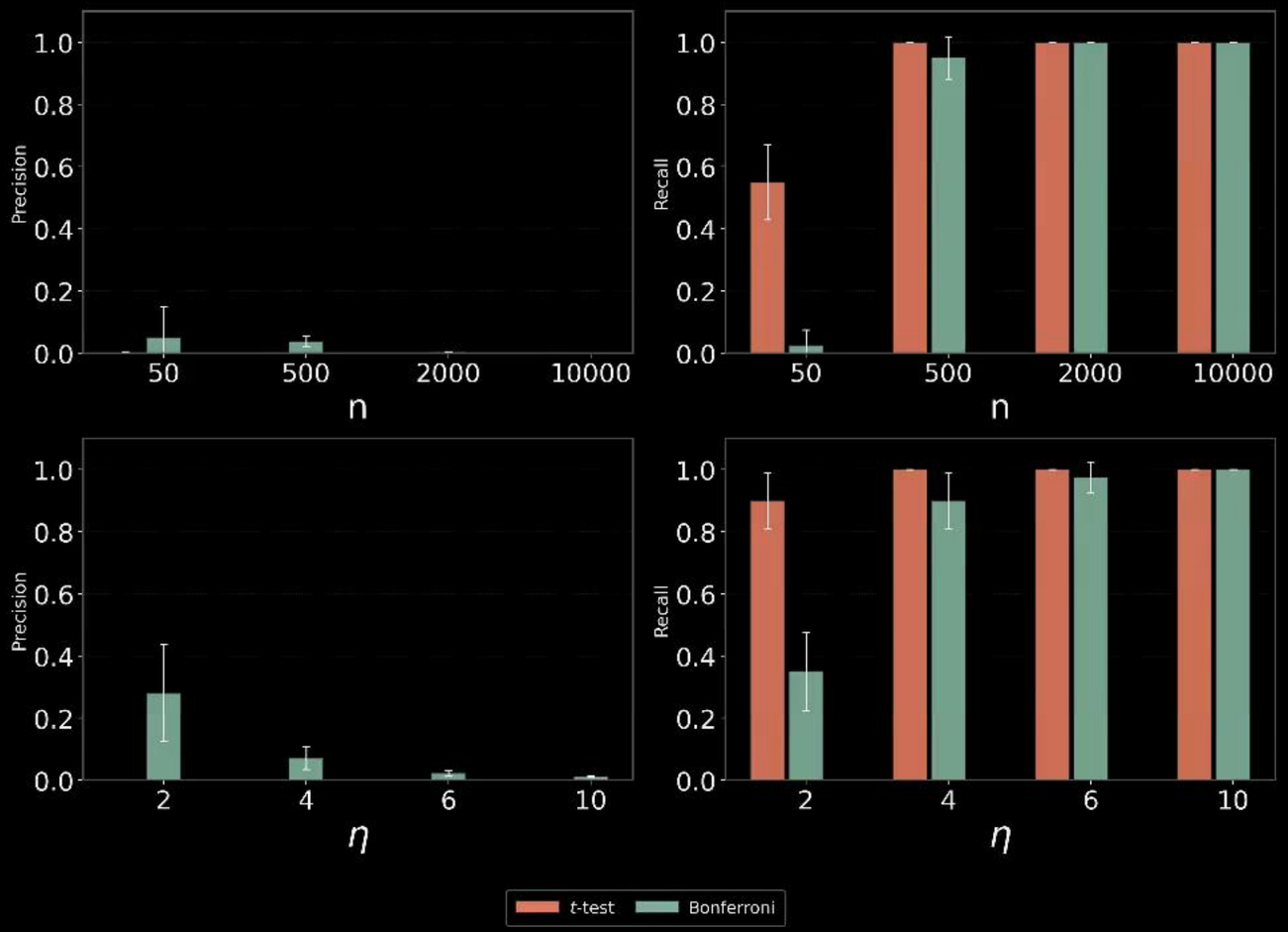
Assumption 2: *Principal Alignment*

=

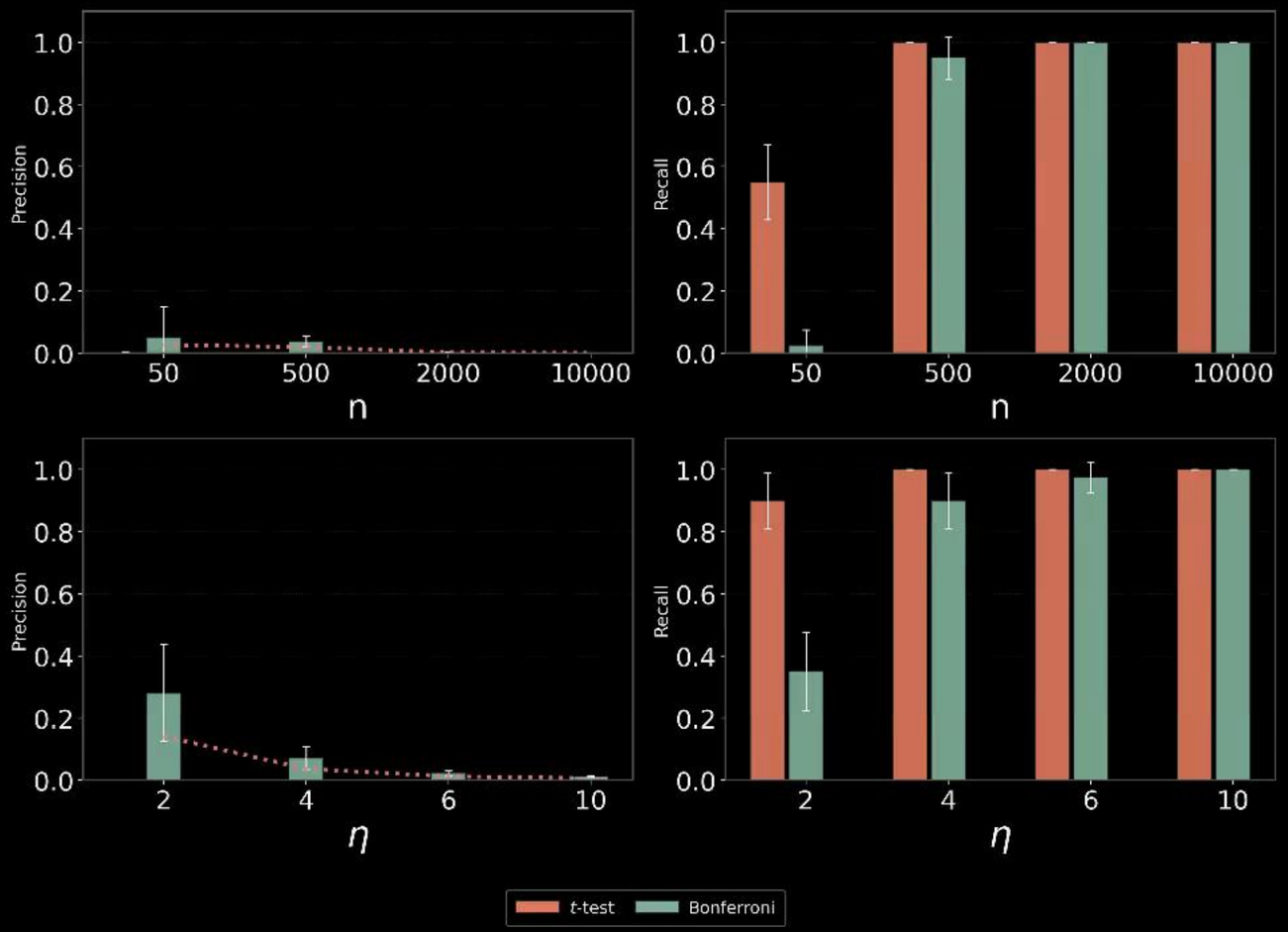


Results: *semi-synthetic (CelebA+)*

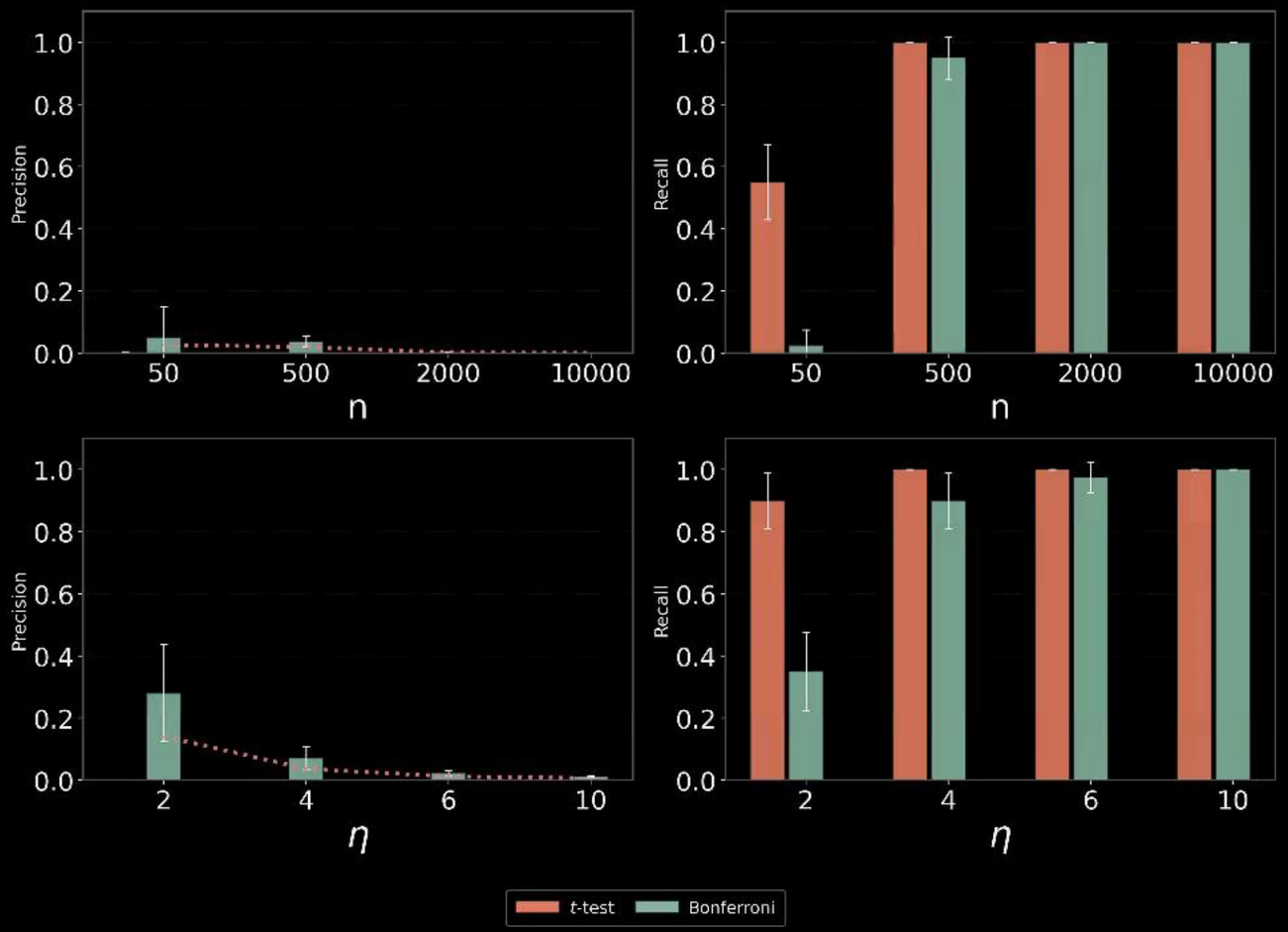
Results: *semi-synthetic (CelebA+)*



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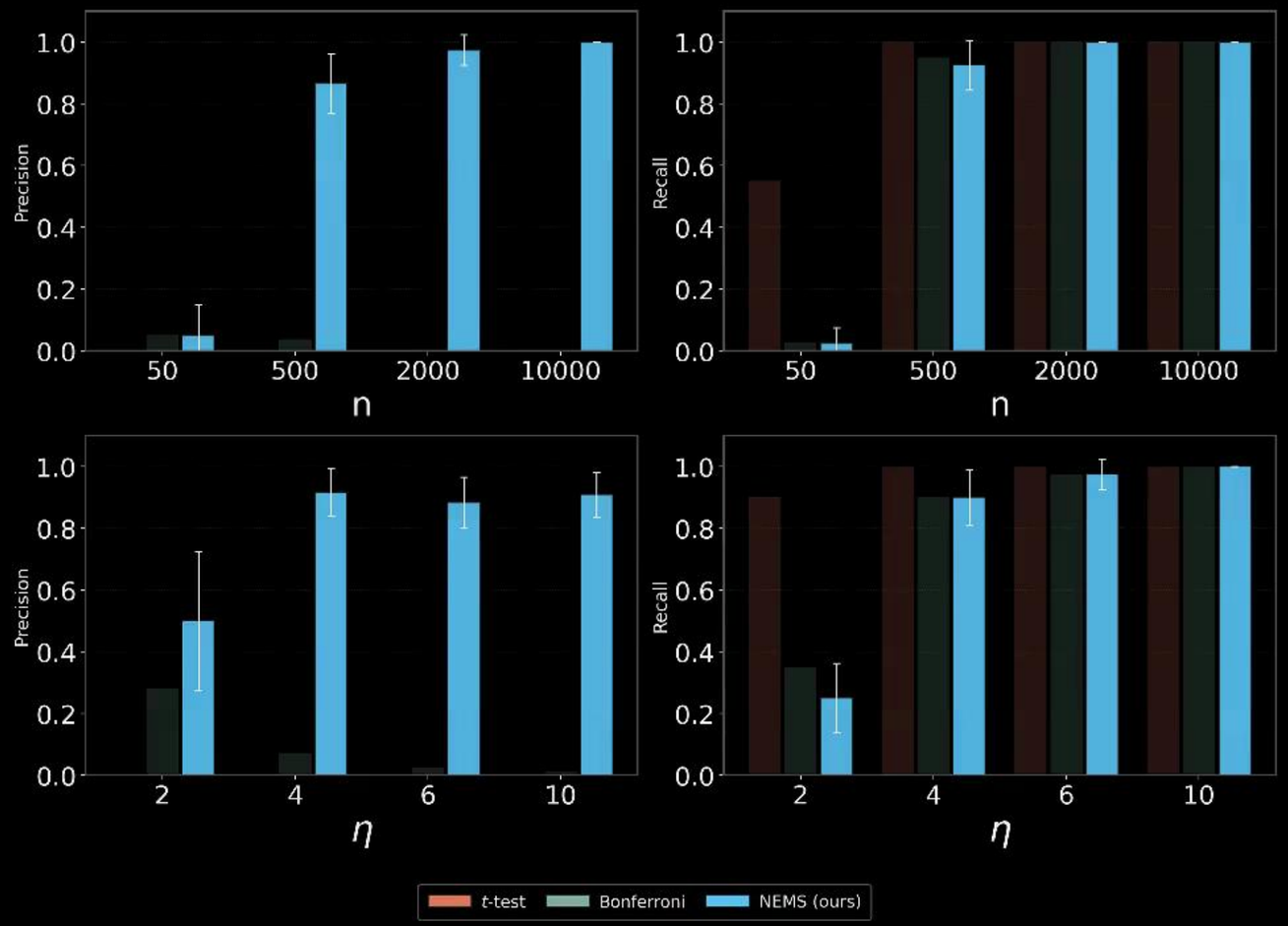


Results: *semi-synthetic (CelebA+)*



Multiple Testing: *precision collapse*

Results: *semi-synthetic (CelebA+)*



Multiple Testing: *precision collapse*

NEXIS (ours): ~~*precision collapse*~~

Results: *Two anti-poverty programs*

See: https://www.riccardocadei.com/NEXIS/



“Measure what is measurable,
and make measurable what is not.”

— Galileo Galilei (1564–1642)

Thank you for your attention

@riccardocadeii